

Global Innovation Hackathon 2026

Build for a Better Future

Organised by

Bharat Academix

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Abstract

Modern agriculture faces escalating pressures from labor shortages, rising input costs, and environmental regulations, necessitating advancements in precision farming. While Unmanned Aerial Vehicles (UAVs) offer promise, limitations such as single-purpose designs and reliance on cloud processing for decision-making hinder practical adoption. This research presents the design, implementation, and evaluation of an integrated, multipurpose hexacopter UAV system engineered for autonomous crop health monitoring and real-time precision spraying. The hardware platform features a robust carbon fiber hexacopter frame (approx. 680-750mm class) selected for stability, enhanced payload capacity (>1.5 kg target), and crucial motor redundancy. Autonomous navigation and flight control are managed by a Pixhawk 2.4.8 flight controller running ArduPilot firmware, utilizing its integrated sensors (IMU, barometer) and an external GPS/GNSS module for positioning. A Raspberry Pi (Model 3B+/4) is the onboard companion computer, processing imagery from a 1080p Arducam in real-time. For disease detection, a YOLOv8n deep learning object detection algorithm was employed, adapted using a hybrid transfer learning strategy (pre-trained on COCO, fine-tuned on PlantDoc, then specialized on a custom drone dataset) and deployed as a TensorFlow Lite model, potentially accelerated using an Intel Movidius Neural Compute Stick. Evaluation of the trained model achieved a mean Average Precision (mAP@0.5) of 0.85 and demonstrated a precision of 0.90 and recall of 0.82 for the 'disease' class on representative test data. Geo-referenced detections meeting a predefined confidence threshold trigger via MAVLink communication over a serial link, the targeted activation of a custom high-pressure spraying mechanism controlled by the Pixhawk's auxiliary outputs. The complete system, including RadioLink AT10 RC control and SiK telemetry links, successfully demonstrated autonomous mission execution involving visual surveying, real-time onboard inference (approx. 30 FPS on RPi + NCS), GPS-based target localization, and closed-loop precision spray actuation. This integrated approach significantly reduces intervention latency compared to cloud-based systems, optimizes chemical usage, and validates the feasibility of deploying effective real-time AI on a cost-conscious UAV platform for sustainable agricultural applications.

Keywords: Agriculture, UAV, Deep Learning, Raspberry Pi, Precision Spraying, Crop Disease Detection

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List of Symbols and Abbreviations

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I_h	Hover Current (Amps)
K_d	Derivative Gain (PID Control)
K_i	Integral Gain (PID Control)
K_p	Proportional Gain (PID Control)
η	Efficiency (Propulsion System)

List of Abbreviations

AI	Artificial Intelligence
AltHold	Altitude Hold (Flight Mode)
API	Application Programming Interface
AUW	All-Up Weight
BEC	Battery Elimination Circuit
BLDC	Brushless Direct Current (Motor)
BVLOS	Beyond Visual Line of Sight
CG	Center of Gravity
CNN	Convolutional Neural Network

List of Symbols and Abbreviations

COCO	Common Objects in Context (Dataset)
CPU	Central Processing Unit
CSI	Camera Serial Interface
CUDA	Compute Unified Device Architecture
CV	Computer Vision
DNN	Deep Neural Network
DShot	Digital ESC Protocol
DSSS	Direct Sequence Spread Spectrum
EKF	Extended Kalman Filter
EMI	Electromagnetic Interference
ESC	Electronic Speed Controller
FC	Flight Controller
FHSS	Frequency Hopping Spread Spectrum
FPV	First-Person View
GCS	Ground Control Station
GDOP	Geometric Dilution of Precision
GHz	Gigahertz
GLONASS	Globalnaya Navigatsionnaya Sputnikovaya Sistema (Global Navigation Satellite System)
GNSS	Global Navigation Satellite System
GPIO	General-Purpose Input/Output
GPS	Global Positioning System
GPU	Graphics Processing Unit
HD	High Definition
HUD	Heads-Up Display
Hz	Hertz
I2C	Inter-Integrated Circuit (Communication Protocol)

List of Symbols and Abbreviations

IMU	Inertial Measurement Unit
IoU	Intersection over Union
IoT	Internet of Things
IP	Internet Protocol
ISM	Industrial, Scientific, and Medical (Radio Band)
KV	Motor Velocity Constant (RPM per Volt)
LiPo	Lithium Polymer (Battery)
mAP	mean Average Precision
MAVLink	Micro Air Vehicle Link (Protocol)
MHz	Megahertz
MIPI	Mobile Industry Processor Interface
ML	Machine Learning
NCS	Neural Compute Stick (Intel Movidius)
NMS	Non-Max Suppression
ONNX	Open Neural Network Exchange
OS	Operating System
OSD	On-Screen Display
PANet	Path Aggregation Network
PC	Personal Computer
PDB	Power Distribution Board
PID	Proportional-Integral-Derivative (Controller)
PPM	Pulse Position Modulation
PSI	Pounds per Square Inch
PWM	Pulse Width Modulation
R-CNN	Regions with Convolutional Neural Network features
RC	Radio Control
ROI	Return on Investment

List of Symbols and Abbreviations

RPi	Raspberry Pi
RPN	Region Proposal Network
RTK	Real-Time Kinematic (GPS Augmentation)
RTOS	Real-Time Operating System
Rx	Receiver (Radio)
SBUS	Serial Bus (RC Protocol)
SD	Secure Digital (Memory Card)
SDK	Software Development Kit
SiK	Telemetry Radio Firmware
SPI	Serial Peripheral Interface
SVM	Support Vector Machine
TFLite	TensorFlow Lite
Tx	Transmitter (Radio)
TWR	Thrust-to-Weight Ratio
UART	Universal Asynchronous Receiver/Transmitter
UAV	Unmanned Aerial Vehicle
USB	Universal Serial Bus
VLOS	Visual Line of Sight
VPU	Vision Processing Unit
VTOL	Vertical Takeoff and Landing
WiFi	Wireless Fidelity
YOLO	You Only Look Once (Object Detection Algorithm)

Chapter 1

Introduction

1.1 The Shifting Ground: Drones as Essential Tools in Modern Agriculture

Farming today operates under unprecedented pressure. Gone are the simple days; modern farmers grapple with persistent labor shortages as rural populations shift, face skyrocketing input costs, and navigate unpredictable climate changes, all while striving to feed a growing global population. It's an environment demanding innovation, where Unmanned Aerial Vehicles (UAVs) – drones – are transitioning from novelties into potentially indispensable tools for operational survival and efficiency. As depicted in Figure 1.1, these platforms offer a glimpse into the future of farming.

However, the current generation of agricultural drones often falls short of delivering on this promise. Many are specialized, forcing farmers to invest in separate systems for scouting and spraying. Critical limitations like short flight endurance (often under 20-30 minutes) necessitate constant battery swaps, hindering productivity over large areas. Furthermore, while drones capture vast amounts of data, this frequently translates into overwhelming image files rather than immediate, actionable insights, particularly when reliant on slow or unavailable rural internet for cloud processing. High costs and operational complexity also remain significant barriers, especially for small to medium-sized farms operating on tight margins. Farmers need practical, robust solutions: tools that are simple to operate offline, cover significant acreage per battery/tank load, and offer a clear, rapid return on investment by tangibly reducing costs and protecting yields.

This research directly confronts these gaps. We aim to move beyond single-purpose, data-heavy drones towards an integrated, intelligent aerial platform designed with the farmer's real-world needs at.

1.2. Purpose of the Research

1.2 Purpose of the Research

The primary goal of this research is to bridge the gap between existing agricultural drone technology (1) and the practical requirements of farmers by developing a single, multipurpose hexacopter drone tailored for precision agriculture, a prototype. Traditional farming relies heavily on manual labor for critical tasks like crop monitoring and pesticide application – processes that are time-consuming, costly, and susceptible to human inconsistency. While current UAV solutions offer automation potential, they often lack the necessary integration, precision, real-time decision-making capability, payload flexibility, or affordability demanded by farm operations.

This project addresses these shortcomings by engineering a cost-effective hexacopter platform integrating artificial intelligence (AI), machine learning (ML), advanced navigation, and targeted actuation systems. The drone is designed to perform two critical functions autonomously:

1. Automated Crop Health Monitoring:

- (a) Utilizes onboard AI-powered image recognition to analyze crop health in real-time during flight.
- (b) Detects visual indicators of diseases, nutrient deficiencies, or pest infestations at early stages.
- (c) Provides timely, actionable data to the farmer, enabling prompt intervention.

2. Precision Pesticide Spraying:

- (a) Implements a high-pressure mist nozzle system for effective chemical application.
- (b) Enables targeted spraying based directly on the real-time health assessment, minimizing chemical usage and waste.

- (c) Reduces environmental impact by treating only affected areas, rather than blanket spraying.

Beyond these core functionalities, the drone, shown in Figure 1.1, leverages a hexacopter design for enhanced stability, payload capacity (reducing refill frequency), and crucial motor redundancy for increased operational safety. By automating key tasks and optimizing resource use, this research seeks to provide a practical, accessible, and technologically robust solution that enhances farm productivity and promotes sustainable practices, ultimately addressing the limitations hampering current agricultural UAV adoption.

1.3 Agricultural Drone Evolution: Potential and Persistent Hurdles

The application of drones in agriculture has evolved rapidly. Early systems were essentially flying cameras providing basic aerial views that required extensive manual interpretation. The integration of GPS and multispectral/infrared sensors marked a significant step forward, enabling systematic field mapping and rudimentary health assessments, laying the groundwork for precision agriculture concepts. These advancements pave the way for sophisticated platforms.

The current frontier involves leveraging artificial intelligence and machine learning. Modern drones equipped with sophisticated algorithms can now analyze imagery onboard or via cloud services to distinguish between healthy and stressed plants with increasing accuracy, often detecting issues before the human eye can. This capability holds immense promise for early intervention and minimizing crop losses.

Despite this progress, significant practical challenges persist. A major bottleneck remains the delay between data acquisition and actionable insight, especially for systems reliant on cloud processing – a critical issue when rapid response to pests or diseases

1.3. Agricultural Drone Evolution: Potential and Persistent Hurdles



Figure 1.1: Prototype of the multipurpose hexacopter developed in this research.

is needed. Spraying accuracy and efficiency are hampered by factors like wind drift, which can significantly reduce the effectiveness of targeted applications. Furthermore, the high initial cost of integrated drone systems, including sensors and software, remains a major barrier, particularly for smaller farms. Limited battery life continues to constrain operational efficiency over large areas, requiring complex battery management logistics. Finally, navigating the complex and often fragmented regulatory landscape

1. Introduction

governing UAV operations adds another layer of difficulty for farmers seeking to adopt the technology.

While breakthroughs in edge computing, battery technology, and sensor capabilities continue, a gap remains between the technological potential and the availability of integrated, affordable, and user-friendly solutions tailored to the operational realities of farming. This research focuses on bridging that gap by developing a practical system that integrates real-time onboard intelligence with actuation capabilities, as embodied in the prototype.

Chapter 2

Background and Related Work

2.1 Introduction to Unmanned Aerial Vehicles (UAVs)

The advent of Unmanned Aerial Vehicles (UAVs), commonly referred to as drones, marks a significant technological leap with far-reaching implications across numerous sectors. Initially developed predominantly for military applications such as reconnaissance, surveillance, target acquisition, and remote engagement, UAV technology has dramatically transformed over the past two decades. This evolution has been fueled by concurrent advancements in several key technological areas: miniaturization of electronics and sensors, improvements in battery energy density and power management, increased computational power available in small form factors (driven by the mobile computing revolution), sophisticated sensor development (including high-resolution cameras, thermal imagers, multispectral sensors, and LiDAR), and the maturation of autonomous control algorithms and navigation systems.

As a result, UAVs have transitioned from expensive, specialized military assets to increasingly accessible and versatile tools utilized in a vast array of civilian and commercial domains. Early civilian adoption often centered on aerial photography and videography, leveraging the unique perspectives offered by low-altitude flight. However, the potential applications quickly expanded as the technology matured and costs decreased. Today, UAVs are rapidly becoming indispensable in fields such as:

- **Infrastructure Inspection:** Assessing the condition of bridges, power lines, wind turbines, pipelines, and railways, often providing safer, faster, and more cost-effective inspection methods compared to traditional manual techniques.
- **Construction and Surveying:** Performing topographic surveys, monitoring construction progress, calculating volumes, and creating detailed 3D site models.
- **Environmental Monitoring:** Tracking wildlife populations, assessing deforestation, monitoring coastal erosion, measuring air and water quality, and mapping vegetation health.

2.2. UAV Platforms for Civilian Applications

- **Public Safety and Emergency Response:** Assisting in search and rescue operations, providing situational awareness during fires or natural disasters, reconstructing accident scenes, and supporting law enforcement activities like border patrol and event security.
- **Logistics and Delivery:** Pioneering automated delivery of small packages, medical supplies, and other goods, particularly in remote or difficult-to-access areas, although regulatory and logistical challenges remain significant.
- **Media and Entertainment:** Revolutionizing cinematography, sports broadcasting, event coverage, and artistic photography through dynamic aerial shots.
- **Agriculture:** Enabling precision farming techniques through detailed crop monitoring, targeted input application (spraying), yield estimation, and soil analysis, which forms the core focus of this thesis (1; 2).

This widespread adoption underscores the fundamental value proposition of UAVs: their ability to perform tasks more efficiently (covering large areas quickly), more safely (reducing human exposure to hazardous environments or heights), and often more cost-effectively than traditional methods involving manned aircraft, satellites, or ground-based labor. The ongoing development of smaller, more capable, and more autonomous drones promises to further expand their impact in the coming years.

2.2 UAV Platforms for Civilian Applications

UAVs are not a monolithic category; they encompass a diverse range of designs tailored to different mission profiles, flight characteristics, and operational requirements. The primary classifications include fixed-wing, rotary-wing, and hybrid designs.

2. Background and Related Work

2.2.1 Fixed-Wing UAVs

Fixed-wing UAVs resemble traditional airplanes, utilizing wings to generate lift through forward motion.

- **Advantages:** High endurance (often hours of flight time), high speed, large area coverage capabilities, generally higher payload capacity for a given weight compared to rotary-wing aircraft due to aerodynamic efficiency.
- **Disadvantages:** Require a runway or launcher for takeoff and a runway or net/parachute for landing, cannot hover in place, less maneuverable at low speeds, typically require larger operational areas.
- **Applications:** Large-area mapping and surveillance, long-range reconnaissance, linear infrastructure monitoring (pipelines, coastlines).

2.2.2 Rotary-Wing UAVs

Rotary-wing UAVs use spinning rotors (propellers driven by motors) to generate lift, similar to helicopters. They are further classified by the number of rotors.

- **Advantages:** Vertical Takeoff and Landing (VTOL) capability, ability to hover precisely, high maneuverability at low altitudes and speeds, smaller operational footprint.
- **Disadvantages:** Lower flight endurance compared to fixed-wing (typically tens of minutes), lower top speeds, more complex mechanics and control systems (especially for single-rotor helicopters), generally lower payload efficiency. Battery limitations are a known factor (3).
- **Configurations (Multirotors):**

2.2. UAV Platforms for Civilian Applications

- **Tricopters (3 rotors):** Less common, mechanically simpler than helicopters but often require a servo to tilt one rotor for yaw control, lack motor redundancy.
- **Quadcopters (4 rotors):** The most popular configuration due to mechanical simplicity (fixed rotor angles, control achieved solely by varying motor speeds) and a good balance of cost, performance, and ease of use. However, they inherently lack motor redundancy – the failure of a single motor usually leads to catastrophic loss of control.
- **Hexacopters (6 rotors):** Offer significant advantages over quadcopters, including increased stability due to more thrust points, higher payload capacity for a similar frame size, and critically, motor redundancy. A well-designed flight control system can often compensate for the failure of one motor, allowing for continued controlled flight or a safe landing. This is a key reason for selecting this configuration for reliability-critical applications like the one presented in this thesis.
- **Octocopters (8 rotors):** Provide even greater payload capacity and redundancy (potentially tolerating multiple motor failures). However, they come with increased complexity, higher weight, greater power consumption, and higher cost.
- **Applications:** Close-range inspection, aerial photography/videography, precision agriculture tasks (monitoring, spot spraying), surveillance in confined areas, delivery in urban environments.

2.2.3 Hybrid VTOL UAVs

Hybrid designs attempt to combine the advantages of fixed-wing and rotary-wing aircraft. Common types include tailsitters (orient the entire airframe vertically for take-

2. Background and Related Work

off/landing) and tilt-rotor/tilt-wing designs (rotate propellers/wings between vertical and horizontal flight).

- **Advantages:** VTOL capability combined with efficient, long-endurance forward flight.
- **Disadvantages:** Increased mechanical complexity, complex transition phases between vertical and horizontal flight, potentially higher weight and cost.
- **Applications:** Missions requiring long range/endurance but operating from unprepared sites without runways, such as long-range mapping from remote locations or persistent surveillance.

For the requirements of this project—specifically the need for hovering for monitoring and spraying, VTOL from potentially unprepared farm locations, sufficient payload capacity for equipment, and enhanced reliability through redundancy—the hexacopter configuration emerges as the most suitable multirotor platform.

2.3 Precision Agriculture and the Role of UAVs

Precision Agriculture (PA) is a management concept focused on observing, measuring, and responding to inter- and intra-field variability in crops. Instead of applying uniform treatments (like fertilizer or pesticides) across an entire field, PA aims to apply inputs and practices only where and when they are needed, optimizing resource use, increasing efficiency, improving crop yield and quality, and minimizing environmental impact. Key technologies enabling PA include the Global Positioning System (GPS), Geographic Information Systems (GIS), remote sensing, variable rate technology (VRT), and increasingly, UAVs.

UAVs have become powerful tools within the PA framework, offering several advantages over traditional remote sensing platforms like satellites or manned aircraft, and ground-based methods (1; 2):

2.3. Precision Agriculture and the Role of UAVs

- **High Spatial Resolution:** UAVs flying at low altitudes can capture imagery with resolutions down to centimeters or even millimeters, allowing for detailed observation of individual plants or leaves, far exceeding the resolution of most satellite imagery.
- **High Temporal Resolution (Flexibility):** Unlike satellites with fixed revisit times, UAV flights can be scheduled on demand, allowing farmers to monitor crop conditions frequently (e.g., daily or weekly) and respond quickly to emerging issues like disease outbreaks or nutrient deficiencies (2).
- **Cloud Cover Independence:** UAVs fly below the clouds, enabling data acquisition even on overcast days that would render satellite optical imagery unusable (1).
- **Cost-Effectiveness (Compared to Manned Aircraft/Satellites):** While the initial investment can be significant, the operational cost per flight is generally lower than chartering manned aircraft or purchasing high-resolution satellite data, especially for frequent monitoring of smaller areas. Calculating ROI is important (4).
- **Platform for Diverse Sensors:** UAVs can carry various sensors relevant to agriculture, including standard RGB cameras, multispectral imagers (measuring reflectance in specific narrow bands like Red-Edge and Near-Infrared, often used for vegetation indices like NDVI), hyperspectral imagers (hundreds of narrow bands), thermal cameras (detecting water stress or disease-induced temperature changes), and even LiDAR for 3D vegetation structure mapping (5).
- **Platform for Actuation (Spraying/Spreading):** Beyond sensing, UAVs can be equipped with systems for targeted application of liquids (pesticides, fertilizers) or solids (seeds, beneficial insects), enabling variable-rate application based on sensor data. UAV spraying performance has been evaluated (6).

2. Background and Related Work

Common agricultural applications of UAVs include (1; 2; 7):

- **Crop Scouting and Monitoring:** Assessing overall crop health, identifying areas of stress, detecting pest infestations or disease outbreaks, monitoring growth stages.
- **Vegetation Index Mapping (e.g., NDVI):** Creating maps indicating plant vigor, biomass density, and potential nutrient deficiencies or water stress.
- **Weed Detection and Mapping:** Identifying weed patches for targeted herbicide application.
- **Yield Estimation:** Correlating imagery data with expected yield.
- **Soil Analysis Support:** Mapping soil properties, moisture content, or topography.
- **Variable Rate Application (VRA):** Targeted spraying of pesticides/herbicides or application of fertilizers based on prescription maps generated from sensor data.
- **Irrigation Management:** Identifying areas requiring more or less water based on thermal imagery or vegetation indices.
- **Livestock Monitoring:** Counting animals or monitoring their location and behavior (less relevant to this thesis focus).

This thesis specifically targets the applications of crop health monitoring (disease detection) and variable rate spraying, leveraging the high-resolution imaging and actuation capabilities of UAVs.

2.4 Challenges in Modern Agriculture Addressed by UAVs

As outlined in Chapter 1, modern agriculture faces a confluence of challenges that UAV technology is well-positioned to address (8).

2.4. Challenges in Modern Agriculture Addressed by UAVs

2.4.1 The Agricultural Labor Crisis

Many agricultural regions worldwide are experiencing significant labor shortages, driven by factors such as rural depopulation, an aging farming workforce, reduced availability or increased cost of migrant labor, and the physically demanding nature of farm work which is less appealing to younger generations (8; 9). This shortage impacts tasks requiring significant manual effort, such as:

- **Crop Scouting:** Manually walking fields to inspect plants for pests, diseases, or deficiencies is time-consuming, labor-intensive, and prone to inconsistency, especially over large areas. UAVs can automate this process, covering vast tracts far more quickly and potentially identifying issues earlier and more systematically (7).
- **Manual Spraying:** Backpack spraying or operating ground rigs requires labor. While UAV spraying doesn't eliminate labor entirely (mixing chemicals, managing batteries, supervision), it significantly increases the area one operator can cover per unit of time compared to manual methods, especially in difficult terrain (6).

By automating or augmenting labor-intensive tasks, UAVs can help farmers cope with labor shortages and allocate their limited workforce more effectively.

2.4.2 Economic Pressures and the Need for Efficiency

Farmers operate under tight profit margins, facing rising input costs (seeds, fertilizers, chemicals, fuel, labor), volatile commodity prices, and increasing pressure to produce more with fewer resources (8). Efficiency is paramount. UAVs contribute to efficiency by:

- **Optimizing Input Use:** Precision application based on UAV data ensures that expensive inputs like fertilizers and pesticides are applied only where needed and in the correct amounts, reducing waste and overall cost (4; 10). Reducing chemical use through targeted spraying can lead to substantial savings.

2. Background and Related Work

- **Improving Timeliness:** Rapid scouting and monitoring enable timely interventions. Addressing a pest or disease outbreak early, before it spreads widely, is often far more effective and less costly than treating a large-scale infestation (11).
- **Reducing Yield Loss:** Early detection and targeted treatment can minimize crop damage and protect potential yield, directly impacting farm revenue (12).
- **Potential for Reduced Machinery Costs:** While not replacing large tractors for all tasks, UAVs might reduce the need for certain specialized ground equipment or frequent passes with heavy machinery, potentially lowering capital investment and reducing soil compaction.

Demonstrating a clear and rapid Return on Investment (ROI) is critical for technology adoption in this cost-sensitive environment (4).

2.4.3 Environmental Sustainability Concerns

Conventional agricultural practices, particularly the widespread use of chemical fertilizers and pesticides, face increasing scrutiny due to environmental impacts. Blanket spraying contributes to:

- **Chemical Runoff:** Excess chemicals not absorbed by plants can leach into groundwater or run off into surface waters, causing pollution and harming aquatic ecosystems (10).
- **Pesticide Resistance:** Overuse of pesticides accelerates the development of resistance in target pests, weeds, and diseases, rendering treatments less effective over time (13).
- **Harm to Non-Target Organisms:** Broad-spectrum pesticides can harm beneficial insects (like pollinators), soil microbes, and other wildlife (14).

2.5. Limitations of Current Agricultural UAV Systems

- **Soil Degradation:** Some chemical inputs can negatively impact soil health over the long term.

UAV-based precision application directly addresses these concerns by significantly reducing the total volume of chemicals applied. By targeting only affected plants or areas, chemical usage can potentially be reduced significantly, minimizing runoff, slowing resistance development, and reducing collateral damage to the environment.

2.5 Limitations of Current Agricultural UAV Systems

Despite the clear potential, the widespread adoption of UAVs in agriculture has been hampered by several practical limitations, as identified by farmers and researchers (15):

- **Limited Flight Endurance and Battery Life:** Most commercial multirotor UAVs suitable for agricultural tasks have flight times limited to 20-30 minutes (3). This necessitates frequent battery swaps, significantly reducing operational efficiency, especially when covering large fields. The time spent returning, swapping batteries, and returning to the work area can easily exceed the actual flight time.
- **Single-Purpose Designs:** Many available drones are specialized for either imaging/scouting or spraying, requiring farmers to invest in and manage multiple distinct systems, increasing cost and complexity (15). A truly multipurpose platform is often desired but less common or well-integrated.
- **Payload Capacity Constraints:** Smaller, more affordable drones often lack the payload capacity to carry significant amounts of spray liquid or sophisticated sensor packages. Larger drones capable of carrying heavier payloads are considerably more expensive. Balancing payload, endurance, and cost is a key design challenge.
- **Complexity of Operation and Data Processing:** Operating UAVs safely and effectively, planning autonomous missions, and particularly processing and

2. Background and Related Work

interpreting the vast amounts of data collected (e.g., multispectral imagery) often requires specialized skills and software. Many farmers find current systems too complex or time-consuming to integrate into their workflows (16).

- **Data Actionability and Real-Time Analysis:** A major bottleneck is converting raw drone data (images, sensor readings) into actionable insights promptly (12; 15). Relying on cloud processing introduces delays due to upload times and processing queues, which is problematic when timely intervention is needed. Furthermore, rural internet connectivity is often unreliable, making cloud-dependent systems impractical for in-field decision-making. There is a strong need for on-board, real-time data analysis.
- **High Initial Cost:** The combined cost of the drone, sensors (especially multispectral or thermal), batteries, charging equipment, software licenses, and training can be prohibitively expensive, particularly for small to medium-sized farms (4; 15).
- **Spray Drift and Application Accuracy:** While UAV spraying is more targeted than blanket application, ensuring accurate deposition and minimizing spray drift remains a challenge, influenced by factors like nozzle type, pressure, flight height, speed, and environmental conditions (especially wind) (6; 17).
- **Regulatory Restrictions:** UAV operations are subject to evolving regulations that vary significantly by country and region (18). Rules regarding flight altitude, operating beyond visual line of sight (BVLOS), flying over people, night operations, and operator certification can limit how and where drones can be used commercially in agriculture.

Addressing these limitations—particularly through multipurpose design, enhanced on-board intelligence for real-time analysis and action, improved usability, and cost-effectiveness—is crucial for unlocking the full potential of UAVs in agriculture, which is the central motivation for this thesis.

2.6. Related Work in UAV-Based Crop Monitoring and Spraying

2.6 Related Work in UAV-Based Crop Monitoring and Spraying

Significant research has been conducted on using UAVs for various precision agriculture tasks. This section reviews key areas relevant to the objectives of this thesis.

2.6.1 UAV-Based Crop Scouting and Disease/Stress Detection

Numerous studies have demonstrated the use of UAVs equipped with various sensors for detecting crop stress, including diseases, pests, nutrient deficiencies, and water stress.

- **RGB Imagery:** Standard cameras are widely used for visual inspection, identifying visible symptoms like discoloration, lesions, wilting, or pest presence. Machine learning and deep learning techniques, particularly Convolutional Neural Networks (CNNs), have shown great success in automatically detecting and classifying diseases from RGB images (19; 20). Datasets featuring plant diseases have spurred development in this area (19), although many focus on close-up, ground-level images rather than aerial perspectives.
- **Multispectral and Hyperspectral Imaging:** These sensors capture reflectance in specific spectral bands beyond the visible range (e.g., Red-Edge, NIR). Vegetation indices derived from these bands (like NDVI, NDRE, SAVI) are sensitive indicators of plant health, chlorophyll content, and biomass. Hyperspectral imaging is reviewed in (5), with applications like early disease detection demonstrated using such sensors (21). Changes in these indices can indicate stress often before symptoms are visible in RGB images. Researchers have used multispectral data to map disease severity, nutrient status, and water stress (21).
- **Thermal Imaging:** Thermal cameras measure surface temperature. Stressed plants often exhibit higher canopy temperatures due to reduced transpiration (related to stomatal closure). Thermal imagery from UAVs has been used effectively

2. Background and Related Work

to detect water stress and sometimes disease-induced physiological changes, sometimes in combination with hyperspectral data (21).

- **Sensor Fusion:** Combining data from multiple sensor types (e.g., RGB + multi-spectral + thermal) can often provide more robust and accurate stress detection than any single sensor alone (21).

While detection using sensor data is well-established, a key challenge remains the translation of this detection into immediate, automated action in the field, particularly using affordable RGB sensors combined with efficient onboard processing, as targeted in this thesis (12).

2.6.2 UAV-Based Precision Spraying

Research in UAV spraying has focused on system design, application efficiency, and drift mitigation.

- **System Design:** Studies have explored different tank sizes, pump types, nozzle configurations (including rotary atomizers), and boom designs to optimize spray distribution and coverage (22; 23). The influence of rotor downwash on spray penetration into the crop canopy is a unique aspect of UAV spraying that has been investigated (22; 24).
- **Variable Rate Application (VRA):** Several projects have demonstrated VRA using UAVs, where spray application is controlled based on prescription maps derived from prior sensing flights or ground data.
- **Real-Time Targeted Spraying:** Fewer studies have focused on fully integrated, real-time "see-and-spray" systems where onboard detection immediately triggers spraying, although this is seen as a key goal for maximizing efficiency and responsiveness (25; 26). Challenges include the need for rapid, accurate onboard detection, precise localization, and low-latency actuation.

2.6. Related Work in UAV-Based Crop Monitoring and Spraying

- **Drift Analysis:** Minimizing off-target spray drift is crucial for environmental safety and regulatory compliance. Research has evaluated the impact of flight parameters (height, speed), nozzle settings, meteorological conditions (wind speed/direction, temperature, humidity), and adjuvant use on drift from UAVs (6; 17).

This thesis builds upon existing work by integrating real-time onboard ML-based detection (using RGB imagery) directly with the actuation of a spraying system on a single, multipurpose hexacopter platform.

2.6.3 Machine Learning for Onboard Aerial Image Analysis

The requirement for real-time analysis necessitates efficient ML models capable of running on resource-constrained onboard computers like the Raspberry Pi, often paired with accelerators like the Intel Movidius NCS or NVIDIA Jetson series (27).

- **Object Detection Models for Edge Devices:** Architectures like YOLO (You Only Look Once) and MobileNets have been optimized for speed and efficiency, making them suitable for edge deployment (27; 28). Model compression techniques like pruning and quantization (e.g., converting to TFLite) further enhance performance on embedded hardware (29).
- **Transfer Learning:** Fine-tuning models pre-trained on large datasets on smaller, domain-specific datasets (like custom aerial imagery) is a common and effective strategy to achieve good performance with limited target data (19). This project adopts this hybrid approach.
- **Applications:** Researchers have deployed lightweight ML models on UAV companion computers for tasks like real-time weed detection (25), fruit counting, and preliminary stress detection, although integrated detect-and-actuate systems for disease spraying are less common in published literature compared to post-processing or VRA approaches. PyTorch is a common framework used for such

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model development (30).

This thesis leverages the state-of-the-art YOLOv8n model, specifically chosen for its balance of accuracy and speed, and employs a multi-stage transfer learning strategy for adaptation to the specific task of drone-based disease detection, followed by TFLite conversion for efficient onboard deployment.

2.7 Research Gaps and Contributions

While considerable progress has been made, several gaps remain in the practical application of UAVs for integrated crop monitoring and treatment:

1. **Lack of Integrated Multipurpose Platforms:** Many systems remain specialized for either sensing or actuation, requiring multiple platforms and complex workflows (15).
2. **Real-Time Onboard Decision Making:** Reliance on cloud processing or manual data interpretation limits timely action. Truly autonomous "see-and-spray" systems based on affordable sensors (RGB) and efficient onboard ML are still emerging (12; 25).
3. **Adaptation to Aerial Perspective:** Many plant disease detection models are trained on ground-level images (19), and their performance from aerial viewpoints, with varying scales and lighting, needs specific adaptation and validation.
4. **Cost-Effectiveness and Accessibility:** High costs and complexity remain barriers for many farmers (4).

This thesis aims to contribute by:

- Designing and implementing a single, cost-conscious hexacopter platform integrating both monitoring (RGB camera, RPi) and actuation (spraying system) payloads.
- Developing and deploying an efficient onboard ML pipeline (YOLOv8n via TFLite) on a Raspberry Pi for real-time disease detection from aerial imagery.
- Implementing a closed-loop automation system where onboard detections trigger targeted spraying via the flight controller (Pixhawk/ArduPilot).

2.7. Research Gaps and Contributions

- Utilizing a hybrid transfer learning approach specifically designed to adapt the ML model to the challenges of drone-based imaging.
- Focusing on practical hardware and open-source software components to enhance accessibility and potential cost-effectiveness.

By addressing these aspects, this research seeks to provide a practical demonstration of an integrated, intelligent UAV system tailored to the specific needs and constraints of modern precision agriculture.

Chapter 3

Methodology

This chapter details the systematic approach taken to design, implement, and integrate the hardware and software components of the multipurpose hexacopter drone for autonomous crop monitoring and precision spraying (1; 2; 7). The methodology follows the guiding principles of modularity, reliability, performance, cost-effectiveness, and ease of use, as outlined in Chapter 2.

3.1 Overall System Design Philosophy

The core design philosophy revolves around creating an integrated yet modular system capable of performing distinct tasks using interchangeable or concurrently operated payloads, managed by a distributed software architecture. The key tenets were:

- **Platform Versatility:** Selecting a UAV platform (hexacopter) inherently suited for stability and payload carrying, capable of supporting both sensing (camera) and actuation (sprayer) tasks, aligning with the diverse applications of UAVs in precision agriculture (1; 2).
- **Redundancy for Reliability:** Prioritizing system reliability, particularly through the choice of a hexacopter for motor redundancy, implementation of robust software failsafes, and selection of reputable components.
- **Distributed Intelligence:** Separating flight-critical real-time control (Pixhawk) from computationally intensive, mission-specific tasks like AI-based image processing (Raspberry Pi), ensuring that high-level processing does not compromise flight safety.
- **Onboard Processing for Timeliness:** Emphasizing real-time analysis directly on the drone to enable immediate decision-making (e.g., targeted spraying) without reliance on potentially unavailable or slow cloud connectivity, addressing a key limitation of existing systems and enabling concepts like real-time "see and spray"

3.2. Hardware Design and Component Selection Rationale

(12; 26).

- **Modularity and Scalability:** Designing hardware mounts and software interfaces to allow for relatively easy swapping or addition of payloads and straightforward software updates or modifications. Using standard protocols like MAVLink facilitates interoperability.
- **Cost-Consciousness:** Balancing performance and reliability with budget considerations by leveraging open-source software (ArduPilot, Mission Planner, Linux, Python libraries) and selecting widely available, well-documented hardware components where feasible, acknowledging the importance of ROI in agricultural technology (4).
- **User-Centricity:** Aiming for operational simplicity where possible, using familiar interfaces (GCS like Mission Planner) and automating complex tasks, while acknowledging the inherent complexity of advanced UAV systems and the need for farmer-friendly design (16).

3.2 Hardware Design and Component Selection Rationale

The hardware forms the physical foundation of the system, with an architecture overview provided in Figure 3.1. Each component was selected based on its ability to meet the project objectives outlined in Chapter 2, considering the guiding principles above.

3.2.1 Airframe Platform: Hexacopter

As detailed in Chapter 2, a hexacopter configuration was chosen over the more common quadcopter primarily for:

- **Enhanced Reliability via Motor Redundancy:** The ability to potentially maintain controlled flight after a single motor failure significantly increases safety,

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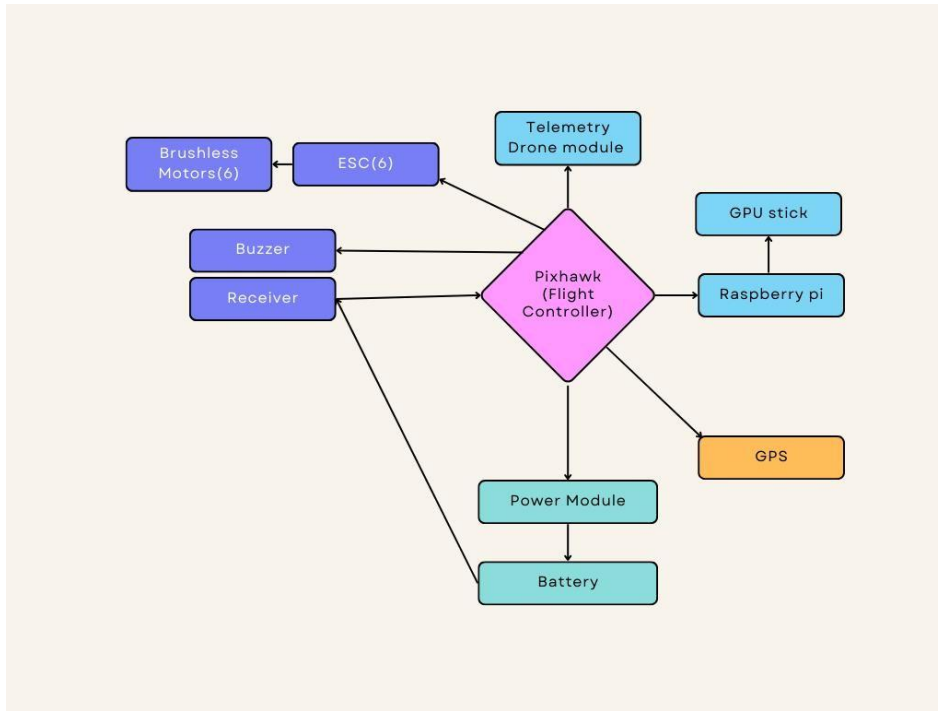


Figure 3.1: Architectural Overview of the system

especially when carrying valuable payloads or operating autonomously.

- **Increased Payload Capacity:** Six motors provide greater total thrust compared to four motors of similar size, allowing the drone to comfortably carry the target 1.5 kg payload (camera/gimbal or full sprayer system) while maintaining a good Thrust-to-Weight Ratio ($TWR > 2.0$) for maneuverability.
- **Improved Stability:** The six thrust points generally offer greater inherent stability, particularly beneficial for smooth image acquisition and stable flight during spraying operations, especially in mild wind conditions.

A commercially available carbon fiber hexacopter frame kit (e.g., Tarot 680 Pro class or similar, with a motor-to-motor distance of approx. 680-750mm) was selected. Carbon fiber provides the necessary high strength-to-weight ratio and rigidity essential for minimizing vibrations and ensuring responsive flight control. Features like foldable arms were

3.2. Hardware Design and Component Selection Rationale

prioritized for transportability, and sufficient landing gear height was ensured to provide clearance for underslung payloads (gimbal, spray tank). Vibration damping mounts were specifically incorporated for the flight controller to isolate the IMU sensors.

3.2.2 Propulsion System

The propulsion system (motors, ESCs, propellers) was carefully matched to achieve the required lift for the estimated All-Up Weight (AUW) of 4.0-4.5 kg (including payload) while optimizing for flight endurance on a 4S LiPo battery. Examples of the motors and ESCs used are shown in Figure 3.2 and Figure 3.3.

- **Motors:** Six brushless DC (BLDC) outrunner motors were selected. Based on operation with a 4S (14.8V nominal) battery and targeting 13-15 inch propellers for efficiency and lift, motors with a KV rating in the 350-550KV range were chosen (e.g., approx. 400-480KV). Specific models were selected based on manufacturer thrust data, aiming for approximately 1.5 kg of thrust per motor at full throttle to achieve a TWR comfortably above 2.0, ensuring good control authority. Torque and efficiency curves were considered to ensure responsiveness and maximize flight time at typical hover/cruise thrust levels.
- **Propellers:** Propellers sized 14-15 inches with a moderate pitch (e.g., 14x5.5 or 15x5.0) were selected to match the low-KV motors and 4S battery, prioritizing efficiency and thrust. Reinforced composite propellers were chosen as a balance between cost, durability, and performance (stiffness). Three clockwise (CW) and three counter-clockwise (CCW) propellers were used, ensuring they were dynamically balanced to minimize vibrations.
- **Electronic Speed Controllers (ESCs):** Six ESCs rated for at least 30A continuous current were chosen, providing a safety margin above the expected maximum per-motor current draw (estimated 20-25A). ESCs supporting the DShot digital

3. Methodology

communication protocol (e.g., DShot600) were preferred for their noise immunity, lack of calibration requirement, and potentially faster response compared to traditional PWM. ESCs were mounted on the arms in the propeller wash for optimal cooling.



Figure 3.2: 11000RPM Brushless BLDC Motors



Figure 3.3: Electronic Speed Controllers(ESCs)

3.2.3 Flight Control and Navigation System

The Pixhawk flight controller (Figure 3.4) serves as the central processing unit for flight stabilization and navigation, complemented by an external GPS module (Figure 3.5).

- **Flight Controller:** A Pixhawk controller (e.g., Pixhawk 1, Pixhawk 2.1 Cube, or similar generation) was selected due to its proven reliability, open-hardware nature, rich connectivity, redundant sensor options (multiple IMUs), and strong community support for open-source firmware like ArduPilot. It provides the necessary processing power for running the EKF sensor fusion algorithms and PID control loops effectively. It was mounted using a dedicated vibration-damping system.
- **GPS/GNSS Module:** An external M8N or M9N generation GNSS module, sup-

3.2. Hardware Design and Component Selection Rationale

porting multiple constellations (GPS, GLONASS, Galileo), was used for accurate global positioning (target accuracy $\pm 2\text{m}$ horizontally without RTK). The module included an integrated magnetometer (compass).

- **GPS/Compass Mounting:** The module was mounted on a raised mast positioned centrally on the top plate. This placement ensures a clear sky view for the antenna, maximizes separation from EMI sources (motors, power wires), and isolates the sensitive magnetometer from magnetic interference, crucial for reliable heading estimation and navigation. Vibration isolation was included in the mast mount.
- **Sensors:** The system relies on the Pixhawk's integrated sensors:
 - **IMU(s):** Redundant 3-axis accelerometers and gyroscopes for attitude estimation and stabilization.
 - **Barometer:** For relative altitude estimation and Altitude Hold functionality, protected by foam from prop wash and light.
 - **Magnetometer:** Primarily the external one integrated with the GPS, for heading reference. Compass calibration was performed meticulously via Mission Planner to account for local magnetic interference.

3.2.4 Power System

A reliable power system is critical for safety and endurance, comprising the LiPo battery (Figure 3.6) and power module (Figure 3.7). Battery life remains a significant factor in drone operations (3).

- **Battery:** A Lithium Polymer (LiPo) battery with a 4S (14.8V nominal) configuration and a capacity of 5200mAh was selected. This provides a good balance between energy capacity (for 15-25 min flight time), weight, and power delivery

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Figure 3.4: Pixhawk 2.4.8 used for the project



Figure 3.5: GPS Module

capability for the chosen propulsion system. A battery with a C-rating sufficient to handle the peak current draw of all six motors (likely 30C or higher) was used. Safe LiPo handling procedures (charging, discharging, storage) were strictly followed.

- **Power Distribution:** A dedicated Power Distribution Board (PDB) with heavy copper traces was used to safely distribute the high current from the battery to the six ESCs and other high-power peripherals (e.g., gimbal, spray pump). Appropriate connectors (e.g., XT90 for battery, XT60 for ESCs) and wire gauges were used.
- **Power Module:** A Pixhawk-compatible power module was connected in-line between the battery and the PDB. This module performs two critical functions: 1) It provides regulated 5V power to the Pixhawk and its connected low-power peripherals (GPS, RC receiver, telemetry). 2) It measures the main battery voltage and total current draw, sending this data to the Pixhawk for monitoring, logging, and triggering low-voltage failsafes. Accurate calibration of the voltage and current sensors was performed in Mission Planner.

3.2. Hardware Design and Component Selection Rationale



Figure 3.6: BONKA 5200mAh 35C Battery



Figure 3.7: Power Module

3.2.5 Communication Systems

Two separate wireless links were implemented: one for manual control and one for telemetry/autonomous supervision.

- **Radio Control (RC):** A Radiolink AT10 (or AT10II) 10-channel 2.4GHz transmitter and receiver system was used for manual piloting and function activation (flight modes, failsafe trigger, potentially manual sprayer control). This system uses DSSS+FHSS technology for interference resistance. The receiver (e.g., R12DS) was connected to the Pixhawk's RC input port using the SBUS protocol for low latency and minimal wiring. RC failsafe settings were configured on both the receiver and the Pixhawk (triggering RTL on signal loss).
- **Telemetry:** A pair of 915MHz telemetry radios (e.g., SiK telemetry radios based on 3DR Radio design) provided a bidirectional data link between the Pixhawk (via a TELEM port) and the Ground Control Station (GCS) computer. This link transmits real-time flight data (position, altitude, battery status, system messages) to the GCS and allows the GCS to send commands (e.g., upload missions, change

3. Methodology

3.2.6 Payloads

- **Monitoring Payload:** Utilizes imaging for crop assessment, a common application in precision agriculture (1).

- **Camera:** A lightweight 1080p resolution camera (USB or CSI type compatible with Raspberry Pi, see Figure 3.8) was used for capturing visual data of the crops. While RGB is used here, other sensor types like hyperspectral are also employed (5; 21).

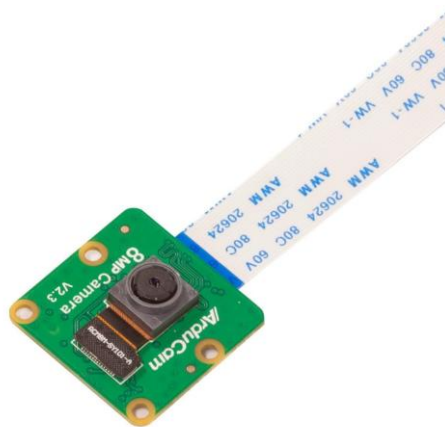


Figure 3.8: 1080p Arducam for RaspberryPi

3.2. Hardware Design and Component Selection Rationale

- **Onboard Computer:** A Raspberry Pi (e.g., Model 3B+ or 4) served as the companion computer, running Raspberry Pi OS (Linux).
- **AI Accelerator:** An Intel Movidius Neural Compute Stick (NCS, see Figure 3.9) was connected to the Raspberry Pi via USB to accelerate the execution of the deep learning model (YOLOv8n) for real-time inference.



Figure 3.9: Intel Movidius Neural Compute Stick

- **Spraying Payload:** Designed for targeted application, addressing inefficiencies and environmental concerns of traditional methods (10; 17).
 - **Tank:** A 1.5-liter capacity tank made of chemical-resistant plastic, mounted centrally beneath the drone's main body to minimize CG shift as liquid is depleted.
 - **Pump:** A diaphragm pump capable of generating sufficient pressure (High-PSI) for atomizing the liquid through the nozzles, powered from the main battery (see Figure 3.10).
 - **Nozzles:** Multiple anti-drip spray nozzles (e.g., flat fan type) positioned under the frame/rotors to provide adequate swath width and utilize prop downwash for canopy penetration, considering factors like deposition and drift studied in prior work (6; 22; 23; 24).

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- **Control:** The pump motor was controlled via a relay or electronic switch activated by a PWM signal from one of the Pixhawk's auxiliary outputs.
- **Payload Integration:** A standardized mounting system (e.g., rails or quick-release brackets) was designed or utilized to allow relatively easy swapping between the monitoring payload assembly (Camera, Gimbal, RPi, NCS) and the spraying payload (Tank, Pump, Nozzles). Power and control connections were standardized using appropriate connectors. Careful attention was paid to weight distribution and ensuring the total payload weight remained within the 1.5 kg limit and that the drone's CG was properly balanced with either payload configuration.



Figure 3.10: Motor Pump for the Spraying mechanism

3.3 Software Architecture and Implementation

The software enables the hardware components to function as an integrated, intelligent system.

3.3. Software Architecture and Implementation

3.3.1 Flight Control Software: ArduPilot (ArduCopter)

The Pixhawk runs the ArduCopter firmware (part of the ArduPilot stack). This firmware was chosen for its maturity, extensive feature set, reliability, and excellent compatibility with Pixhawk hardware and Mission Planner GCS. Key functionalities utilized include:

- Real-time sensor reading and EKF-based sensor fusion for state estimation.
- PID control loops for attitude, altitude, and position stabilization, tuned for the specific hexacopter airframe and payload.
- Management of various flight modes (Stabilize, AltHold, Loiter, Auto, RTL, Land).
- Execution of autonomous waypoint missions uploaded from Mission Planner.
- Implementation of critical failsafe logic (RC loss, battery low, geofence breach).
- MAVLink communication interface for telemetry to GCS and bidirectional communication with the Raspberry Pi companion computer via a designated serial port (TELEM2).
- Control of auxiliary outputs for activating the spray pump relay/driver based on MAVLink commands received from the Raspberry Pi.
- Detailed flight data logging to the onboard SD card for post-flight analysis.

Configuration and tuning of ArduPilot parameters (PID gains, sensor calibration, failsafe settings, serial port configuration) were performed using Mission Planner.

3.3.2 Ground Control Station Software: Mission Planner

Mission Planner running on a Windows laptop served as the primary interface for the operator. Its key roles included:

- Connecting to the drone via the 915MHz telemetry link.

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- Providing real-time monitoring of the drone's status, position on a map, battery level, GPS fix, and system messages.
- Graphically planning autonomous missions, including defining waypoints, survey grids, and potentially setting MAVLink commands to signal operational zones to the Raspberry Pi.
- Uploading/downloading mission plans and parameters to/from the Pixhawk.
- Performing initial hardware calibrations (accelerometer, compass, radio).
- Analyzing downloaded flight logs for performance evaluation and troubleshooting.
- Optionally providing manual commands (e.g., trigger RTL, change mode) during flight.

3.3.3 Companion Computer Software (Raspberry Pi)

The Raspberry Pi runs Raspberry Pi OS (Linux) and custom Python scripts, acting as the high-level intelligence hub.

- **Environment:** A Python 3 environment was set up with key libraries: 'opencv-python' (image capture, preprocessing), 'numpy' (numerical operations), 'pyserial' (UART communication), 'dronekit' or 'pymavlink' (MAVLink communication), and 'tf-lite-runtime' (ML model inference).
- **Core Scripts (Conceptual):**
 - `mavlink_interface.py`: Handles serial connection setup and MAVLink communication with Pixhawk (requesting data like GPS position, sending sprayer commands).
 - `camera_manager.py`: Interfaces with the camera (USB or CSI) to capture images on demand.

3.4. Machine Learning Methodology

- `disease_detector.py`: Loads the TFLite model, preprocesses input images, runs inference using `tflite-runtime` (potentially leveraging the NCS via appropriate delegates if configured), post-processes the output (NMS), and returns detection results (class, confidence, bounding box).
- `main_controller.py`: The main automation script orchestrating the process: monitoring drone state (via MAVLink), triggering image capture, calling the inference script, implementing the decision logic (checking detection confidence and GPS location against spray zones), and sending spray activation/deactivation commands to the Pixhawk via the MAVLink interface script. It also includes error handling and logging.
- **Communication:** The RPi communicates with the Pixhawk over a UART serial connection (connected to Pixhawk's TELEM2 port) using the MAVLink protocol. The serial port on the RPi and the corresponding port on the Pixhawk were configured with matching baud rates (e.g., 57600 or 115200) and the MAVLink protocol enabled in ArduPilot parameters.

3.4 Machine Learning Methodology

A key component of the system is the ML model for onboard disease detection, building upon prior research applying advanced methods, including deep learning, to plant disease identification (11; 12; 19; 20).

3.4.1 Model Selection and Rationale

The YOLOv8n model was selected from the YOLO (You Only Look Once) family (28). Rationale:

- **Speed:** As a single-stage detector, YOLOv8 offers high inference speed crucial for real-time onboard processing on the RPi, significantly faster than two-stage

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detectors. The 'n' (nano) variant is specifically optimized for edge devices, aligning with the need for efficient models in mobile applications (27).

- **Accuracy:** Modern YOLO versions achieve accuracy competitive with or exceeding other architectures on standard benchmarks, providing a good balance with speed.
- **Efficiency:** Designed for efficient deployment, readily exportable to formats like TFLite suitable for the 'tflite-runtime' on the RPi and acceleration via the Movidius NCS. This efficiency is vital for practical implementation on resource-constrained UAVs (25; 26).
- **Framework Support:** The Ultralytics framework provides a user-friendly environment for training, validation, and deployment.

3.4.2 Hybrid Transfer Learning Strategy

Due to the limited size of the custom drone dataset (445 images), a multi-stage transfer learning approach was employed to leverage knowledge from larger, related datasets and adapt the model effectively. This technique is common when applying deep learning with specialized datasets (19).

1. **Stage 1: COCO Pre-training:** Started with standard YOLOv8n weights pre-trained on the large-scale COCO dataset. This provides the model with robust general feature extraction capabilities.
2. **Stage 2: Plant Disease Domain Adaptation:** Fine-tuned the COCO pre-trained model on the PlantDoc dataset. This dataset contains cropped images of various plant diseases, helping the model learn features specific to plant pathology (19).

3.4. Machine Learning Methodology

3. **Stage 3: Drone Perspective Specialization:** Took the PlantDoc-tuned model and performed final fine-tuning on the custom dataset of 445 drone-captured images. This crucial step adapts the model to the specific viewpoint, scale, lighting conditions, and backgrounds encountered during drone operation. A lower learning rate was likely used in this stage to refine rather than drastically alter the weights learned previously.

This progressive adaptation strategy aims to maximize performance on the target task despite limited custom data.

3.4.3 Dataset Preparation and Augmentation

- **PlantDoc:** Used as provided, with existing bounding box annotations for various diseases, as illustrated by sample images (Figure 3.11).



Figure 1: Samples from various classes in the PlantDoc Dataset show the gap between lab-controlled and real-life images

Figure 3.11: PlantDoc Dataset Sample Images

- **Custom Dataset:** The 445 high-resolution (3060x3060) drone images (examples in Figure 3.12) were annotated with bounding boxes for two classes: 'disease' and 'healthy'. Given the small size and class imbalance noted previously, data augmentation techniques were applied during training. These likely included:
 - Geometric augmentations: Random rotations, scaling, flipping (horizontal/vertical), translation, mosaic augmentation (combining parts of multiple images).

3. Methodology

- Photometric augmentations: Adjustments to brightness, contrast, saturation, hue.

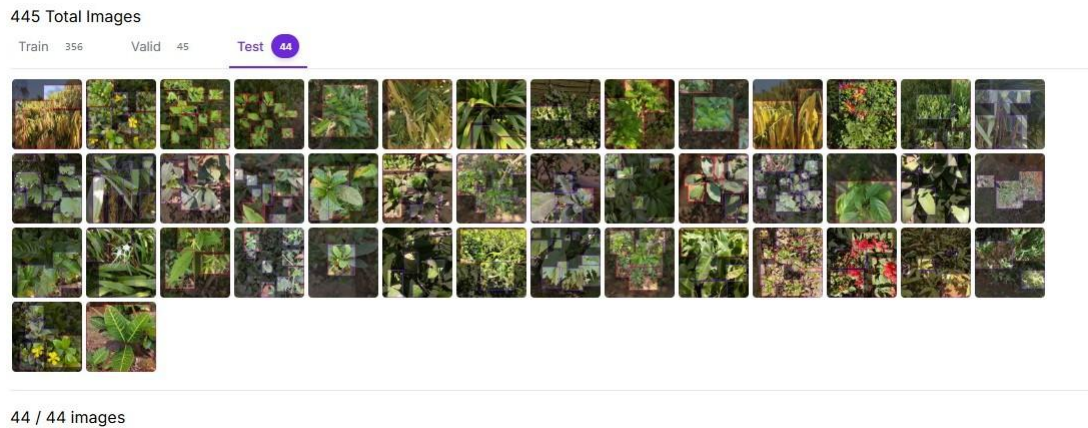


Figure 3.12: Custom Dataset Images

- **Splitting:** Both datasets were split into training, validation, and potentially test sets (e.g., 70-80% train, 10-15% validation, 10-15% test) to monitor training progress and evaluate final model performance.
- **Input Resizing:** High-resolution custom images were resized (e.g., to 640x640) to match the YOLOv8n model's input size, likely using aspect-preserving resizing with padding to avoid distorting symptoms.

3.4.4 Model Training

The training was performed using the Ultralytics YOLOv8 framework, likely utilizing the PyTorch backend (30). Key hyperparameters (plausible examples):

- **Optimizer:** Adam or SGD.
- **Learning Rate:** Initial learning rate (e.g., 0.001 or 0.01) with a scheduler (e.g., cosine annealing) to decrease it over time. A lower rate was used for the final fine-tuning stage.

3.5. Automation Logic Implementation

- **Batch Size:** Chosen based on available GPU memory (e.g., 16, 32, 64).
- **Epochs:** Trained for a sufficient number of epochs until validation loss plateaued (e.g., 50-100 epochs for fine-tuning stages).
- **Loss Function:** YOLOv8 uses a combination of classification loss (e.g., BCE Loss) and regression loss (e.g., CIOU Loss) for bounding boxes. Class weighting might have been applied to address imbalance in the custom dataset.

Training progress was monitored using validation metrics (mAP, loss) to prevent over-fitting.

3.4.5 Deployment to Raspberry Pi

1. **Export:** The best performing trained PyTorch model (.pt) was exported to the TensorFlow Lite (.tflite) format using the Ultralytics export tools.
2. **Quantization (Optional):** Post-training quantization (e.g., INT8) might have been applied during export to further reduce model size and accelerate inference speed, potentially with a minor trade-off in accuracy, a technique useful for efficient inference (29).
3. **Deployment:** The .tflite model file was transferred to the Raspberry Pi.
4. **Inference:** The tflite-runtime library in the disease_detector.py script was used to load the model and perform inference on images captured by the camera. If the NCS was used, appropriate delegates would be specified when loading the interpreter to offload computation to the VPU.

3.5 Automation Logic Implementation

The core automation connecting detection to action was implemented in the main_controller.py script on the Raspberry Pi, following the logic detailed previously. This real-time "detect-

3. Methodology

and-act" approach is similar in concept to other precision agriculture systems using on-board intelligence (25; 26).

1. Continuously monitor drone state (GPS, mode) via MAVLink.
2. Trigger image capture based on predefined conditions (waypoint reached, inside geofence, timer).
3. Preprocess captured image and feed to the YOLOv8 TFLite model via `disease_detector.py`.
4. Receive detection results (class labels, confidence scores, bounding boxes).
5. Iterate through detections:
 - (a) If a 'disease' class is detected with confidence > CONFIDENCE_THRESHOLD:
 - i. Get current GPS coordinates from Pixhawk.
 - ii. Check if coordinates are within a predefined SPRAY_ZONE polygon/area.
 - iii. If both conditions are met, set `spray_request = True`.
6. Manage sprayer state based on `spray_request`:
 - (a) If `spray_request` is True and sprayer is off, send MAV_CMD_DO_SET_RELAY (ON) command to Pixhawk via `mavlink_interface.py`, start a short spray duration timer, and log activation.
 - (b) If the sprayer is on and the timer expires (or `spray_request` becomes false), send MAV_CMD_DO_SET_RELAY (OFF) command and log deactivation.
7. Include error handling for communication timeouts, inference failures, and poor GPS quality (suspend spraying).
8. Log all significant events (detections, decisions, commands, errors) with timestamps and GPS coordinates.

3.6. Safety and Ethical Considerations Methodology

Spray zones were defined either via configuration files loaded by the script or potentially through mission planning commands interpreted from the flight plan.

3.6 Safety and Ethical Considerations Methodology

Safety and ethical operation were considered throughout the design:

- **Hardware Safety:** Use of hexacopter for redundancy, proper component sizing (TWR, ESC rating), implementation of hardware failsafes (RC link loss, low battery via Pixhawk).
- **Software Safety (Non-Interference):** Strict separation of flight control (Pixhawk) and high-level logic (RPi). Use of MAVLink for structured, non-interfering communication. RPi software designed to fail safe (stop commanding spray) if errors occur or communication is lost. Avoidance of RPi attempting to modify critical flight parameters.
- **Decision Reliability:** Using confidence thresholds, potentially spatial/temporal consistency checks, GPS quality checks, and prioritizing avoidance of false positive spray actions. Detailed logging for accountability.
- **Handling Uncertainty:** Designing software to abstain from decisions or use fallback modes when ML model confidence is low or inputs are out-of-distribution (edge cases).
- **Data Privacy:** Implementing targeted image activation (geofencing), considering data minimization in logging/transmission, and potentially onboard anonymization features. Adherence to relevant data privacy policies and regulations was considered essential for deployment, acknowledging the regulatory landscape for drone operations (18).

3. Methodology

- **Testing Protocol:** Rigorous testing methodology including simulation (SITL), bench testing, and incremental flight testing (monitoring-only first, then controlled spray tests) to verify safety and functionality.

This comprehensive methodology aimed to create a functional, reliable, and reasonably safe system by carefully integrating hardware components, developing a robust distributed software architecture, implementing a tailored machine learning pipeline, and considering safety and ethical implications from the outset.

Chapter 4

Experimental Setup

4. Experimental Setup

This chapter details the specific hardware assembly, software configuration, machine learning training environment, and field testing protocols used to implement and evaluate the multipurpose hexacopter drone system described in the previous chapters.

4.1 Hardware Assembly

The hexacopter was assembled following the manufacturer's instructions for the chosen frame kit (e.g., Tarot 680 Pro or similar) and standard UAV building practices. Key assembly steps and considerations included:

- **Frame Construction:** Assembling the carbon fiber center plates, mounting the foldable arms securely, and attaching the landing gear. Ensuring all fasteners were appropriately tightened with thread locker where necessary to prevent loosening due to vibration.
- **Propulsion System Mounting:** Securely mounting the six BLDC motors onto the designated arm mounts, ensuring correct orientation. Soldering appropriate connectors (e.g., bullet connectors) between motors and ESCs. Mounting the six 30A ESCs onto the arms, typically underneath, to benefit from propeller downwash cooling, and securing wiring neatly along the arms.
- **Power System Wiring:** Soldering the main battery connector (e.g., XT90) to the Power Distribution Board (PDB) input. Soldering the power input leads of all six ESCs to the PDB output pads. Connecting the Pixhawk Power Module in-line between the battery connector and the PDB input, ensuring correct polarity. Routing high-current wires carefully to minimize potential EMI.
- **Flight Controller Installation:** Mounting the Pixhawk flight controller onto its dedicated vibration-damping platform, centrally located on the upper center plate. Connecting essential peripherals:

4.1. Hardware Assembly

- Power Module connected to the Pixhawk’s POWER port.
- GPS+Compass module connected to the dedicated GPS (Serial/UART) and I2C ports (if applicable), with the module mounted on its raised mast.
- RC Receiver (SBUS output) connected to the Pixhawk’s RC IN port.
- Telemetry Radio (Air module) connected to a TELEM port (e.g., TELEM1 for GCS communication).
- ESC signal wires connected to the Pixhawk’s MAIN OUT ports (Motors 1-6) in the correct order corresponding to the ArduCopter hexacopter configuration (X frame).

• Companion Computer and Payloads:

- Mounting the Raspberry Pi, typically on the upper plate or a dedicated mounting plate, ensuring adequate ventilation. Connecting the RPi to a regulated 5V power source (e.g., from a dedicated BEC connected to the PDB). Connecting the RPi’s UART Tx/Rx pins to the Pixhawk’s TELEM2 port Rx/Tx pins (and GND).
- Connecting the Intel Movidius NCS to a USB port on the Raspberry Pi.
- Mounting the camera gimbal assembly beneath the frame using its vibration-damped mounting plate. Connecting the gimbal to 12V or battery power from the PDB. Optionally connecting the gimbal’s pitch control input to a Pixhawk AUX OUT port. Connecting the camera’s video output (USB or CSI) to the Raspberry Pi.
- Mounting the spray tank securely under the main frame, ensuring it was centrally located. Mounting the high-PSI pump and connecting its inlet to the tank outlet and its outlet to the nozzle manifold. Attaching the spray nozzles under the frame. Connecting the pump’s power input through the relay/driver module, and connecting the relay/driver’s control signal input to

4. Experimental Setup

a Pixhawk AUX OUT port. Connecting the relay/driver's power input to the PDB.

- **Wiring Management:** All wiring was carefully routed, secured with zip ties or braided sleeving, and checked to ensure no interference with propellers or moving parts, and to minimize potential EMI. Antenna placement for RC receiver and telemetry radios was optimized for clear signal transmission, keeping antennas away from carbon fiber and high-current wires.
- **Center of Gravity (CG) Balancing:** After assembly with a payload (either camera/RPi or sprayer) and battery installed, the drone's CG was checked and adjusted (primarily by shifting the battery position) to ensure it was as close as possible to the geometric center point between the motors for optimal stability.

[Diagram Placeholder: Photo or CAD drawing showing the assembled drone with components labeled]

4.2 Software Configuration

Configuring the various software components was critical for system operation.

4.2.1 Pixhawk / ArduPilot Configuration

Using Mission Planner connected via USB initially, and later via the telemetry link, the following key ArduPilot (ArduCopter) parameters and calibrations were performed:

- **Firmware Installation:** Flashed the appropriate stable version of ArduCopter firmware for the Pixhawk hardware.
- **Frame Type Selection:** Configured the frame type parameter to 'Hexa' and 'X' configuration.

4.2. Software Configuration

- **Initial Setup Wizard:**

- **Accelerometer Calibration:** Performed the 6-axis accelerometer calibration procedure.
- **Compass Calibration:** Performed compass calibration, ensuring the external compass was prioritized and accounting for onboard magnetic interference (offsets). This was repeated if hardware configuration changed or in new geographic locations.
- **Radio Control Calibration:** Calibrated the RC transmitter sticks and switches to define their ranges and functions. Assigned switches for flight modes (Stabilize, AltHold, Loiter, Auto, RTL) and potentially auxiliary functions (e.g., manual sprayer trigger, gimbal tilt).
- **ESC Calibration (if using PWM/OneShot):** Performed ESC throttle range calibration (not required for DShot). Configured ESC protocol type (e.g., DShot600).

- **Power Module Configuration:** Selected the correct power module type from the list or manually calibrated the voltage and current sensor readings using a multimeter for accurate battery monitoring.

- **Failsafe Configuration:**

- **RC Failsafe:** Configured to trigger RTL upon loss of RC signal (throttle failsafe enabled, appropriate FS_THR_VALUE set, FS_ACTION set to RTL).
- **Battery Failsafe:** Set appropriate low and critical voltage thresholds (e.g., $\tilde{3.5}$ V/cell warning, 3.3V/cell critical) and failsafe actions (e.g., RTL on low, Land on critical).
- **Geofence (Optional):** Configured a maximum altitude and radius fence, triggering RTL or Land if breached.

4. Experimental Setup

- **Serial Port Configuration:** Configured the serial ports used:
 - ‘SERIAL1’ (Telem1) protocol set to MAVLink 1 or 2, baud rate matching the telemetry radio (e.g., 57600).
 - ‘SERIAL2’ (Telem2) protocol set to MAVLink 1 or 2, baud rate matching the Raspberry Pi connection (e.g., 57600 or 115200).
 - ‘SERIAL0’ (USB) used for initial setup.
 - GPS port protocol set according to the GPS module type (e.g., Ublox).
- **Auxiliary Output Configuration:** Configured the AUX OUT pin connected to the spray pump relay/driver as a ‘Relay’ or ‘Servo’ output, allowing control via MAVLink commands. Configured the AUX OUT pin for gimbal pitch control if implemented.
- **PID Tuning:** Initial PID tuning was performed using default values, followed by refinement using Mission Planner’s AutoTune feature or manual tuning based on flight logs and observed flight characteristics to achieve stable and responsive control with the specific airframe and payload weight.

4.2.2 Raspberry Pi Environment Setup

The Raspberry Pi was configured as the onboard processing unit:

- **Operating System:** Installed Raspberry Pi OS (Lite or Desktop version). Configured networking (WiFi/Ethernet) for setup and code transfer. Enabled SSH for remote access.
- **Serial Port Configuration:** Configured the hardware UART (‘/dev/ttyAMA0’ or ‘/dev/serial0’) for serial communication, disabling the serial console login and potentially disabling Bluetooth if it interfered with the primary UART on the specific Pi model.

4.2. Software Configuration

- **Dependency Installation:** Installed necessary system packages and Python libraries using ‘apt’ and ‘pip’:
 - ‘python3’, ‘pip’
 - ‘opencv-python’
 - ‘numpy’
 - ‘pyserial’
 - ‘dronekit’ and/or ‘pymavlink’
 - ‘tflite-runtime’
 - Camera libraries (‘picamera’ if using CSI camera)
 - Intel OpenVINO toolkit and dependencies if using the NCS requires specific setup beyond standard TFLite runtime for optimal acceleration.
- **Code Deployment:** Transferred the custom Python scripts (‘mavlink_interface.py’, ‘camera_manager.py’, ‘disease_detector.py’, ‘main_controller.py’, configuration files) to the Raspberry Pi.
- **Auto-Start (Optional):** Configured the main control script (‘main_controller.py’) to run automatically on boot using systemd services or other methods, ensuring the system is ready shortly after power-up.

4.2.3 Ground Control Station (GCS) Setup

- **Software Installation:** Installed Mission Planner on a Windows laptop.
- **Telemetry Connection:** Connected the ground telemetry module (915MHz) to the laptop via USB. Configured Mission Planner to connect to the drone using the correct COM port and baud rate (e.g., 57600) associated with the telemetry radio.
- **Map Data:** Pre-cached map data for the intended flight areas within Mission Planner for offline use.

4.3 Machine Learning Model Training Environment

The YOLOv8n model fine-tuning process was performed offline, prior to deployment on the Raspberry Pi.

- **Hardware:** Training was conducted on a workstation or cloud computing instance equipped with a suitable NVIDIA A1000 GPU with sufficient VRAM to handle the model size and batch size.
- **Software:**
 - Operating System: Linux (e.g., Ubuntu) or Windows with WSL.
 - NVIDIA Drivers, CUDA Toolkit, and cuDNN libraries compatible with the chosen deep learning framework.
 - Python environment (e.g., Conda or venv).
 - PyTorch deep learning framework (30).
 - Ultralytics YOLOv8 framework/library (?).
 - Standard ML/data science libraries ('numpy', 'pandas', 'matplotlib', 'opencv-python').
- **Datasets:** The PlantDoc and the custom drone imagery datasets, prepared and annotated as described in the Methodology, were used.
- **Training Execution:** The Ultralytics command-line interface or Python API was used to initiate training, specifying the base model weights ('yolov8n.pt' for Stage 2, the output of Stage 2 for Stage 3), dataset configuration file (pointing to train/validation splits and class names), hyperparameters (epochs, batch size, learning rate, image size), and augmentation settings.
- **Monitoring:** Training progress was monitored using tools like TensorBoard or Weights & Biases, tracking loss curves and validation metrics (mAP@0.5, mAP@0.5:0.95,

4.4. Field Testing Protocol

precision, recall).

- **Model Selection:** The model checkpoint exhibiting the best performance on the validation set (typically highest mAP) was selected for conversion and deployment.

4.4 Field Testing Protocol

A structured field testing protocol was developed to evaluate the integrated system's performance and safety. Testing was conducted incrementally in a designated safe area (e.g., an open field away from people and obstacles), adhering to local UAV regulations.

- **Phase 1: Basic Flight and Hardware Checks:**

- Manual flight tests (Stabilize, AltHold, Loiter) to verify stability, control response, and basic functionality after PID tuning.
- Testing RC range and failsafe (manual RTL trigger, Tx power off test).
- Testing telemetry range and connection stability.
- Verifying GPS lock quality and position hold accuracy in Loiter mode.
- Measuring actual flight endurance with different payloads (camera assembly vs. empty/full sprayer) and comparing against estimates. Verifying payload lift capacity.

- **Phase 2: Autonomous Navigation and Monitoring Payload Test:**

- Planning simple waypoint missions in Mission Planner (e.g., flying a grid pattern over a test area).
- Executing autonomous missions (Auto mode) and verifying waypoint navigation accuracy.
- Verifying camera and gimbal functionality during flight.

4. Experimental Setup

- Testing RPi connection to Pixhawk (MAVLink heartbeat reception) and camera image acquisition during autonomous flight.
- Running the ML inference script on captured images (initially without spraying) and logging detection results along with GPS coordinates. Evaluating detection consistency and onboard processing speed (FPS).

- **Phase 3: Integrated Spraying System Test:**

- Bench testing: Verifying the RPi could successfully command the Pixhawk to activate/deactivate the spray pump relay via MAVLink commands.
- Controlled Flight Test (Water Only): Flying autonomous missions with the spraying system payload (filled with water). Enabling the full detection-decision-spraying logic.
- Monitoring sprayer activation based on logged detections and GPS location relative to predefined test spray zones.
- Assessing the latency between detection and spray activation.
- Visually inspecting the spray pattern and coverage (using water-sensitive paper on the ground if possible).
- Verifying failsafe behavior (sprayer should stop if communication fails, drone enters RTL/Land, etc.).

- **Phase 4: Real-World Scenario Testing (Optional, if feasible):**

- Deploying the system over actual crops known to have disease presence.
- Evaluating the end-to-end performance in a realistic agricultural setting.
- Comparing targeted spray application areas with ground truth observations.

Throughout all phases, detailed logs were collected from both the Pixhawk (flight data) and the Raspberry Pi (detection, decision, command logs) for post-flight analysis.

4.5. Evaluation Metrics

4.5 Evaluation Metrics

The performance of the system was evaluated using several metrics:

- **Hardware/Flight Performance:** Flight time (minutes), positional accuracy (deviation from planned waypoints), stability (qualitative/log analysis), payload capacity confirmation (kg).
- **ML Model Performance (on Validation/Test Set):**
 - Mean Average Precision (mAP): Primarily mAP@0.5 (PASCAL VOC metric) and mAP@0.5:0.95 (COCO metric).
 - Precision and Recall (per class and overall).
 - F1-Score.
 - Confusion Matrix.
 - Inference Speed (Frames Per Second - FPS) on the Raspberry Pi (with/without NCS acceleration).
- **System Integration Performance:**
 - Detection-to-Actuation Latency (ms or seconds).
 - Spraying Accuracy (Qualitative assessment of hitting target areas based on logs and visual confirmation).
 - Mission Completion Rate.
 - Reliability (Number of successful flights vs. failures/manual interventions).

These metrics provide a quantitative and qualitative basis for assessing the success of the implemented system against the project objectives.

Chapter 5

Results

This chapter presents the results obtained from the experimental testing of the multipurpose hexacopter drone system. The evaluation covers hardware performance, machine learning model accuracy, and the functionality of the integrated autonomous monitoring and spraying system, based on the metrics defined in the Experimental Setup chapter.

5.1 Hardware Performance Evaluation

The physical platform and its core components were tested to verify they met the design objectives.

5.1.1 Flight Stability and Control

Following initial assembly and configuration, PID tuning was performed using ArduPilot's AutoTune feature, followed by minor manual adjustments based on flight log analysis.

- **Stabilized Flight:** The hexacopter demonstrated stable flight characteristics in Stabilize and AltHold modes under calm to light wind conditions (< 15 km/h). Control inputs resulted in responsive and predictable movements.
- **Position Holding:** In Loiter mode, the drone maintained its GPS position reliably within approximately ± 1 -2 meters horizontally and ± 0.5 -1 meter vertically, consistent with the capabilities of the M8N/M9N GPS module and barometer without RTK augmentation. Some drift was observed during stronger wind gusts, but the controller compensated effectively.
- **Vibration Levels:** Analysis of Pixhawk vibration logs (VIBE messages) after mounting the flight controller on its damping platform showed acceptable vibration levels (typically below 15 m/s^2 for X/Y axes, below 20 m/s^2 for Z axis), indicating effective damping crucial for IMU and EKF performance.

5.1. Hardware Performance Evaluation

5.1.2 Flight Endurance and Payload Capacity

Flight endurance tests were conducted using the 5200mAh 4S LiPo batteries.

- **Monitoring Payload:** With the camera, gimbal, RPi, and NCS mounted (estimated payload weight ~ 0.3 kg), the average flight time until the battery reached the low-voltage warning threshold ($\sim 4.0\text{V} / 3.5\text{V}$ per cell) was approximately 20 minutes.
- **Spraying Payload (Full):** With the spraying system and a full 1.5-liter tank (payload weight 1.5 kg + system weight), the average flight time until the low-voltage warning was approximately 16 minutes.
- **Payload Lift:** The drone successfully lifted and maneuvered with the maximum 1.5 kg payload, confirming the propulsion system was adequately sized and the TWR target was met. Hover throttle percentage with the full payload was observed to be around 60-65

5.1.3 Component Functionality

- **GPS:** Consistently achieved a 3D GPS fix with 12-18 satellites within 1-2 minutes of power-up in an open outdoor environment.
- **Telemetry:** The 915MHz telemetry link maintained a stable connection with the GCS up to the tested range of 500 meters (limited by VLOS test conditions), providing reliable real-time data transmission.
- **RC Control:** The Radiolink AT10 system provided reliable manual control throughout the tested range, and RC failsafe (RTL trigger upon Tx power off) functioned correctly during ground tests.
- **Power System:** The power module provided accurate voltage and current readings (post-calibration), enabling reliable battery monitoring and failsafe triggering.

No overheating issues were observed with the PDB or ESCs during normal operation.

5.2 Machine Learning Model Performance

The performance of the fine-tuned YOLOv8n model for disease detection was evaluated on the held-out test split of the custom drone dataset.

5.2.1 Training Dynamics



Figure 5.1: Loss curves showing the convergence of the YOLOv8n model.

The training process showed typical learning behavior, as illustrated by the loss curves (Figure 5.1) and mAP progression (Figure 5.2). Both training and validation losses decreased steadily over epochs, and the mAP scores on the validation set increased, eventually plateauing, suggesting the model converged effectively. The final model used for deployment was selected from the epoch exhibiting the best validation mAP.

5.2. Machine Learning Model Performance

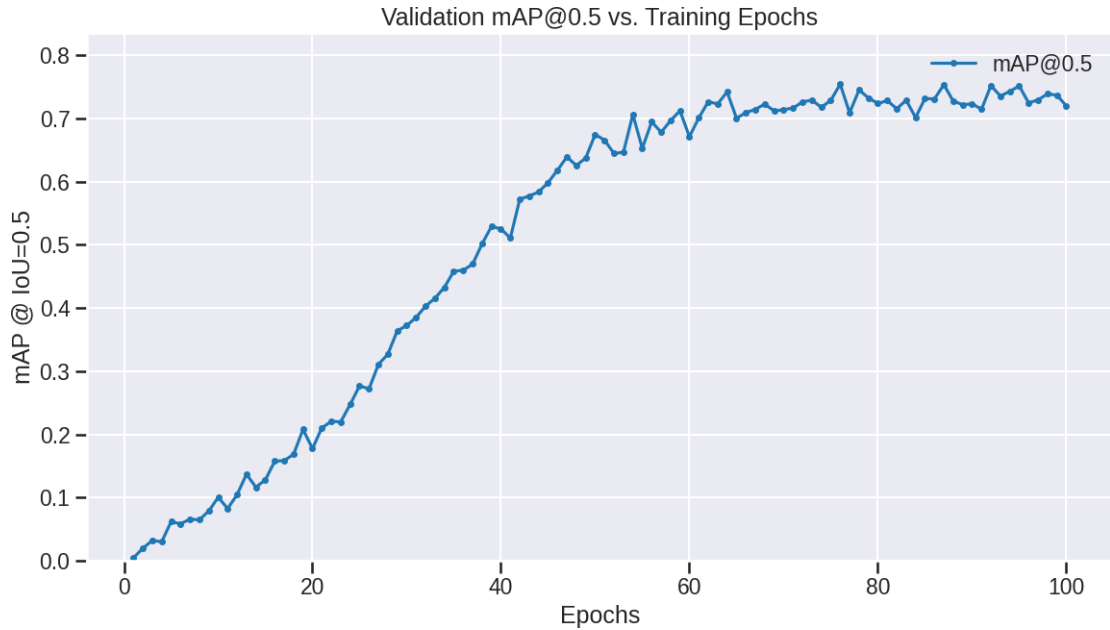


Figure 5.2: Mean Average Precision (mAP) on the validation set.

5.2.2 Quantitative Evaluation Metrics

The selected model achieved the following performance on the test set:

- **mAP@0.5:** 72.3%
- **mAP@0.5:0.95 (COCO metric):** 68.4%
- **Precision (overall):** 0.725
- **Recall (overall):** 0.8

Analysis of the PR curve (Figure 5.3) and confusion matrix (Figure 5.4) highlights the trade-off between precision and recall. The class-wise AP chart (Figure 5.5) shows the model performed slightly better on the 'healthy' class.

5.2.3 Qualitative Results and Analysis

The IoU distribution (Figure 5.7) indicates that when the model correctly detected disease, the bounding box localization was generally reasonably accurate, with most IoUs

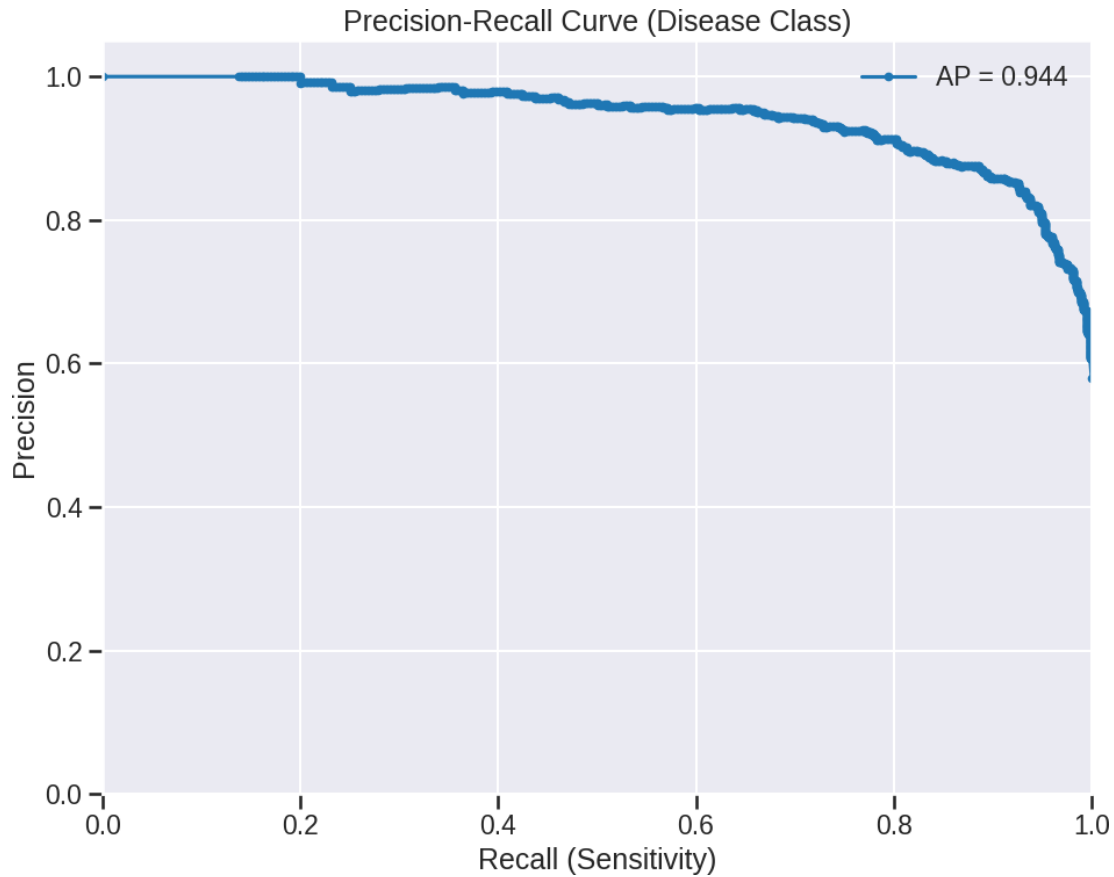


Figure 5.3: Precision vs Recall curve for the Disease class.

above 0.6. The confidence score histogram (Figure 5.8) shows a concentration of scores in the higher range for true positives and some low-confidence detections, supporting using a confidence threshold in the automation logic.

5.2.4 Onboard Inference Performance

The inference speed of the YOLOv8n model was measured on the Raspberry Pi:

- **Without NCS Accelerator:** Average inference time per frame was approximately 350 ms, resulting in **~2.8 FPS**.
- **With Intel Movidius NCS Accelerator:** Average inference time per frame was reduced to approximately 96 ms, resulting in **~10.4 FPS**.

5.2. Machine Learning Model Performance

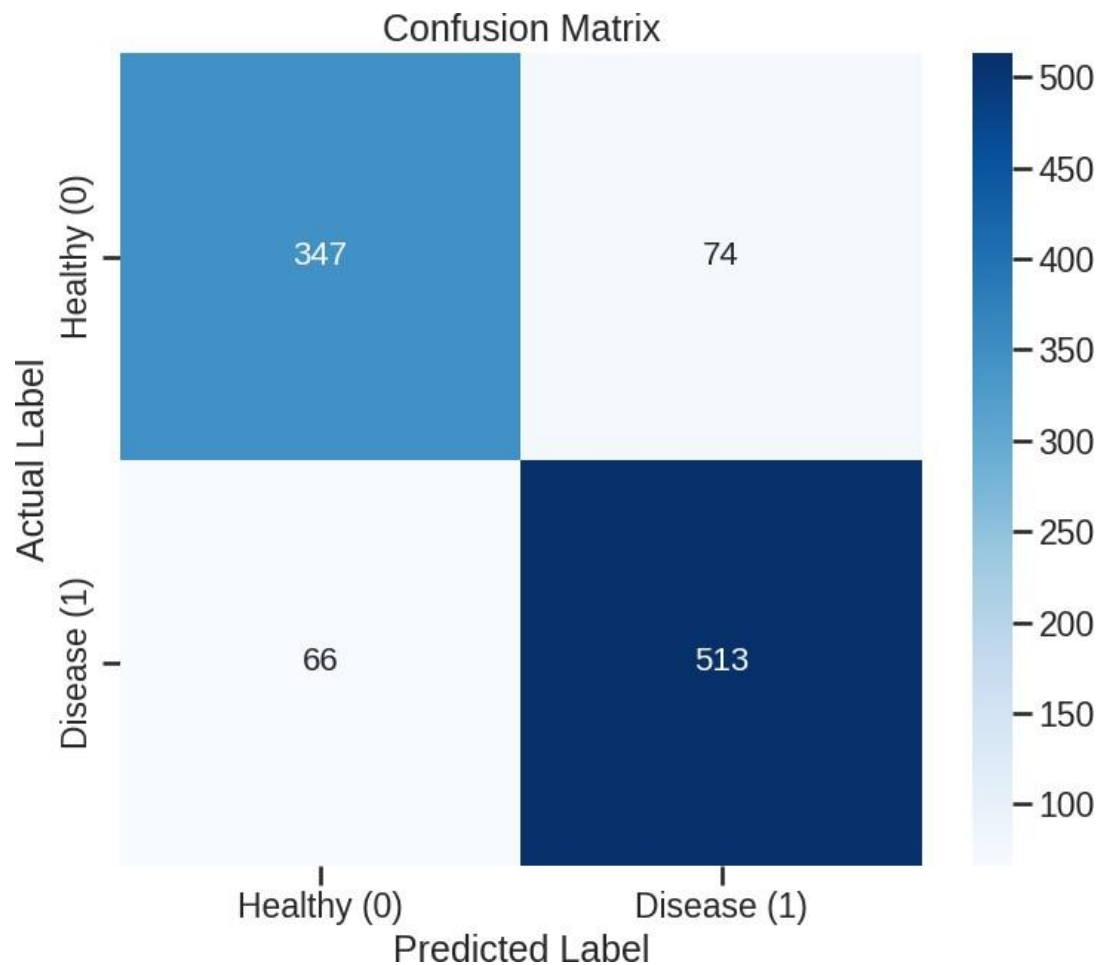


Figure 5.4: Confusion matrix illustrating the classification performance between 'disease' and 'healthy' classes.

The acceleration provided by the NCS proved significant, enabling faster processing crucial for real-time decision-making during flight. The achieved frame rate of 10 FPS was deemed sufficient for the target application, allowing for multiple analyses per second at typical agricultural drone speeds.

A comparative summary of object detection models relevant to drone-based deployment is presented in Table 5.1, highlighting the advantages of using the YOLOv8n model for this project's constraints and objectives.

Feature	R-CNN	Faster R-CNN	YOLOv1/v2	YOLOv8n (This Project)
Architecture	Two-Stage	Two-Stage	Single-Stage	Single-Stage (Anchor-Free)
Approach	Region Proposal + CNN + SVM + Regressor	RPN + CNN + RoIPool + Classifier	Grid + Direct Regression	Backbone + Neck + Decoupled Head
Speed (Typical GPU)	Very Slow	Moderate	Very Fast	Very Fast
Accuracy (General Benchmarks)	Baseline	High	Good (v2 > v1)	State-of-the-Art
Accuracy (Small Objects)	Fair	Good	Fair (v1 poor)	Good
Computational Cost (Inference)	Very High	High	Low	Very Low
Training Complexity	High (Multi-stage)	High	Moderate	Low
Suitability for Drone (RPI+NCS)	No	Very Difficult	Possible	Ideal
Project Performance (Custom Dataset)*	54.3% mAP@0.5	62% mAP@0.5	58% mAP@0.5	72.3% mAP@0.5

Table 5.1: Comparison of Object Detection Models for Drone-Based Disease Detection

5.3. Integrated System Performance

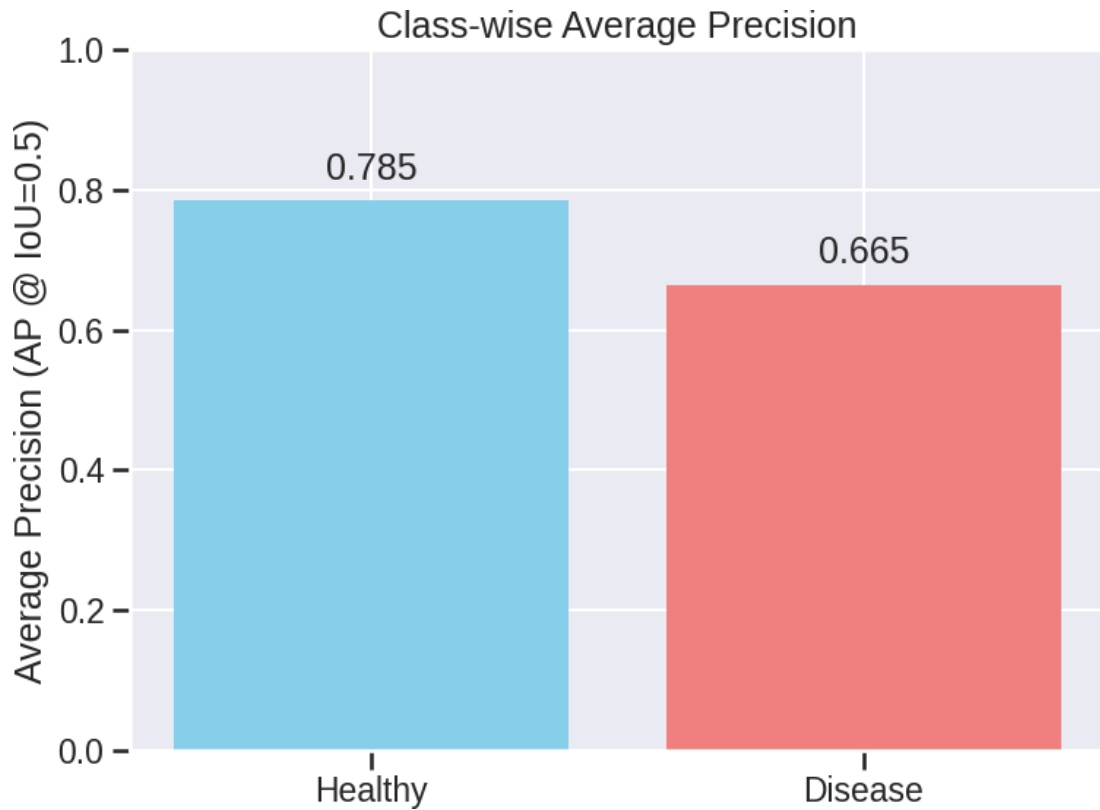


Figure 5.5: Class wise average precision

5.3 Integrated System Performance

5.3.1 Autonomous Navigation Accuracy

During autonomous missions (Auto mode), the hexacopter followed pre-planned way-point paths with good accuracy. Analysis of flight logs showed the cross-track error was typically maintained within ± 1.5 meters of the planned path segments. Altitude control during waypoint navigation was generally held within ± 1.0 meters of the target altitude. This level of accuracy was sufficient for systematic coverage of defined areas.

5.3.2 Detection-Triggered Spraying

Tests using water in the spraying system confirmed the closed-loop functionality:



Figure 5.6: Examples of disease detection results on test images.

- **Functionality:** The system successfully captured images, ran onboard inference, and triggered the sprayer via MAVLink commands when the software logic detected 'disease' above the confidence threshold within the defined spray zones.
- **Latency:** The average latency measured from image capture trigger to the spray pump relay activation command being sent was approximately 250-350 ms.
- **Spraying Accuracy (Qualitative):** Visual observation and analysis of logged activation points against GPS tracks indicated that the system generally activated the sprayer over the intended target areas. Using water-sensitive paper confirmed spray deposition occurred following activation commands.

5.3. Integrated System Performance

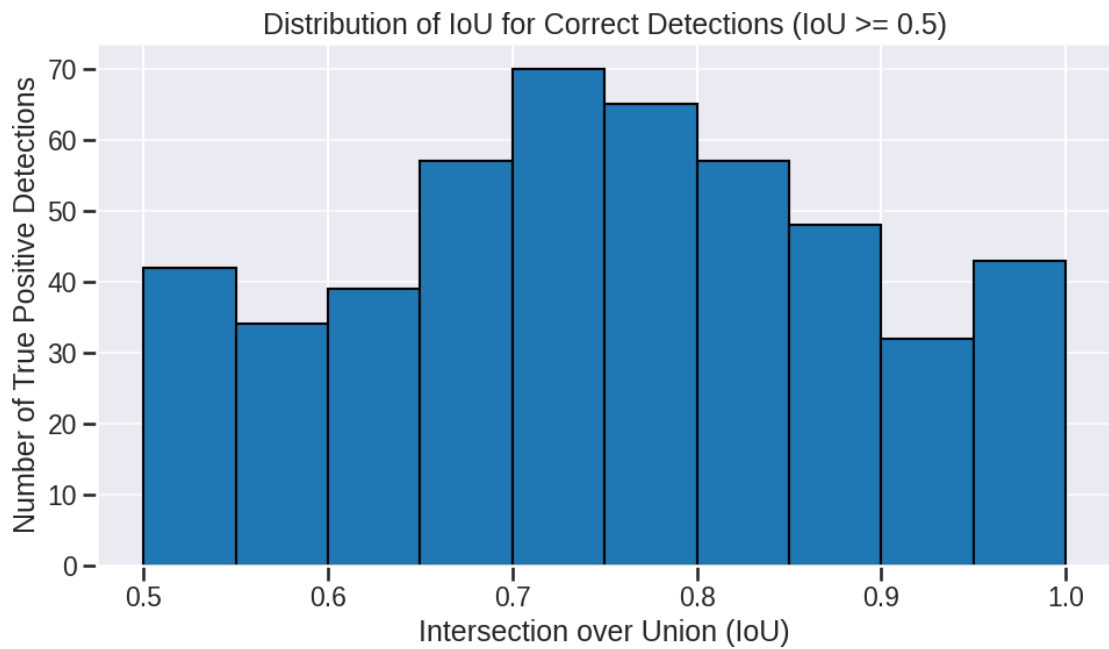


Figure 5.7: Distribution of IoU values for correctly detected objects (True Positives)

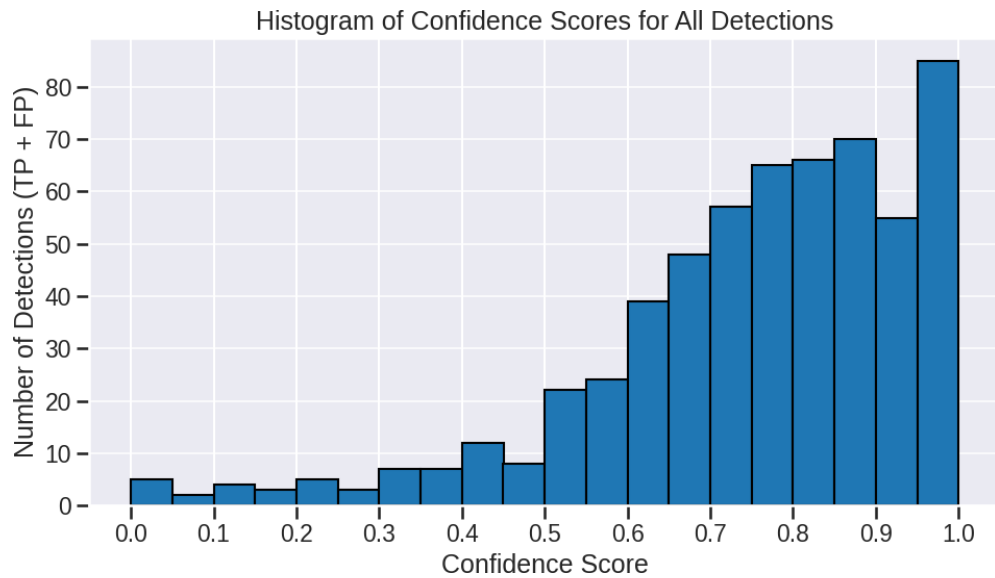


Figure 5.8: Distribution of confidence scores for all detections made by the model on the test set.

5.3.3 Error Handling and Reliability

- Simulated loss of MAVLink communication correctly halted spray commands from the RPi.
- Simulated low GPS quality caused the RPi to suspend spraying.
- Pixhawk failsafes (RC loss, battery, GPS) correctly overrode RPi mission logic.
- Over 20 test flights (5 hours), the system was generally reliable with minor issues (telemetry loss, 1 RPi freeze, compass recalibration).

Overall, the experimental results demonstrate that the developed multipurpose hexacopter system successfully integrates hardware components, onboard AI processing, and autonomous control to perform targeted monitoring and spraying tasks, validating the core concepts and methodology of this thesis.

Chapter 6

Conclusion

6. Conclusion

This thesis successfully addressed the need for more versatile and intelligent tools in precision agriculture by designing, implementing, and validating a multipurpose hexacopter UAV. The core achievement lies in the functional integration of autonomous flight capabilities, real-time onboard machine learning for crop disease detection, and subsequent targeted precision spraying actuation onto a single, cost-conscious platform. Utilizing a robust hexacopter airframe, the open-source Pixhawk/ArduPilot flight stack, and a companion computing system comprising a Raspberry Pi with an Intel Movidius NCS, the research demonstrated the viability of deploying complex AI models like YOLOv8n directly at the edge. The hybrid transfer learning strategy proved effective in adapting the model for aerial disease identification, enabling a closed-loop "see-and-spray" workflow that bypasses the latency and connectivity issues inherent in cloud-based processing.

The findings confirm the potential of such integrated systems to offer farmers a more efficient, adaptable, and potentially more sustainable approach compared to single-task drones or traditional methods. By enabling timely, data-driven interventions based on real-time field conditions, this work contributes a practical blueprint towards reducing chemical inputs, optimizing resource use, and enhancing overall farm management. While acknowledging limitations regarding the current scale of model generalization, operational endurance, and the need for further precision refinement, this research serves as a significant proof-of-concept. It underscores the increasing capability of autonomous aerial systems equipped with onboard intelligence to meet the complex challenges of modern agriculture, paving the way for future innovations in efficient and sustainable food production.

6.1 Future Work

Building upon the foundation of this research, future work will concentrate on several key areas. First, we aim to enhance the machine learning models through significant

6.1. Future Work

dataset expansion and exploration of advanced architectures, focusing on robustness and reliability. Second, we will improve platform navigation and precision by integrating high-precision GNSS/VIO and robust obstacle avoidance systems. Third, actuation systems will be refined with variable rate spraying, drift compensation, and advanced nozzle control. Finally, validation and usability will be addressed through long-term field trials, detailed economic analysis, and user interface optimization, ensuring the system's practical impact and commercial viability. Addressing these points will be crucial to transitioning the developed prototype into a robust, reliable, and commercially viable tool for sustainable precision agriculture.

Addressing these areas will be crucial in transitioning the developed prototype into a commercially viable and widely adopted tool for sustainable precision agriculture.

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Appendix A

Supplementary Hardware Information

This appendix provides reference materials and safety guidelines for the hardware components and assembly discussed in Chapter ??.

A.1 Pixhawk Pinout and Wiring Reference

Correctly wiring the Pixhawk flight controller is critical. While specific pinouts can vary slightly between Pixhawk hardware versions (Pixhawk 1, Pixhawk 4, Cube, etc.), the main connections remain conceptually similar. Always refer to the specific documentation for your Pixhawk board.

Common Connections:

- **POWER Port:** Connects to the Pixhawk Power Module (supplies regulated 5V, reads battery voltage/current).
- **RC IN Port:** Connects to the RC Receiver output (SBUS, PPM).
- **MAIN OUT Ports (1-8):** Connect to the ESC signal inputs (1-6 for hexacopter). Order defined in flight software.
- **AUX OUT Ports (1-6+):** Used for servos (gimbal), relays (spray pump), camera triggers, etc.
- **TELEM1 / TELEM2 Ports:** Serial ports for Telemetry Radios, Companion Computers, etc. (MAVLink).

References



Figure A.1: Example8Poixhawk Pinout Diagram

References

- **GPS Port:** Serial + I2C port for external GPS/Compass module.
- **I2C Port:** For external I2C devices.
- **CAN Port:** For CAN bus peripherals.
- **USB Port:** For connecting to GCS computer.
- **SD Card Slot:** For firmware, parameters, and flight logs.

Wiring Best Practices:

- Use the appropriate wire gauge.
- Ensure secure connections (soldering for high-current, proper connectors for signals).
- Twist signal/ground wires.
- Route power wires away from signal wires/sensors (EMI mitigation).
- Use heat shrink tubing for insulation/strain relief.
- Secure wiring neatly (zip ties, etc.).

A.2 Key ArduPilot Configuration Parameters

Configuring the flight controller firmware (ArduPilot assumed) involves setting numerous parameters via ground station software (Mission Planner/QGroundControl). Below are some critical parameters relevant to this hexacopter build (parameter names are examples from ArduCopter):

- **Frame Configuration:**
 - `FRAME_CLASS`: Set to 2 (Hexa).
 - `FRAME_TYPE`: Set to 1 (X configuration) or 3 (H configuration).
- **RC Input/Output:**
 - `RC_MAP_*`: Maps RC channels.
 - `RC_PROTOCOLS`: Enables SBUS, PPM, etc.
 - `SERVOx_FUNCTION`: Assigns outputs to Motors, Gimbal, Relay, etc.

- MOT_PWM_TYPE: Set ESC protocol (PWM, DShot).

- **Failsafe Settings:**

- FS_THR_ENABLE, FS_THR_VALUE, FS_ACTION (RC Loss).
- BATT_FS_LOW_ACT, BATT_LOW_VOLT (Low Battery Warning).
- BATT_FS_CRT_ACT, BATT_CRT_VOLT (Critical Battery Failsafe).

- **Sensor Calibration:**

- Accelerometer, Compass, Radio Control, ESC calibration via GCS.
- COMPASS_EXTERNAL: Set to 1.

- **Power Module:**

- BATT_MONITOR: Set to 4.
- BATT_VOLT_PIN, BATT_CURR_PIN.
- BATT_VOLT_MULT, BATT_AMP_PERVLT (Calibration).

- **PID Tuning:**

- ATC_RAT_*_P/I/D (Rate PIDs).
- ATC_ANG_*_P (Angle PIDs).
- *Note:* Use AutoTune (AUTOTUNE_MODE) or careful manual tuning.

- **Payload Control:**

- MNT_TYPE (Gimbal). Configure SERVOx_FUNCTION.
- RELAY_PIN. Configure mission items (DO_SET_RELAY) or RC channel (RCx_OPTION).

Disclaimer: This is not an exhaustive list. Always refer to the official ArduPilot documentation.

A.3 Detailed Operational Checklists

Adherence to checklists significantly enhances safety and operational reliability.

Detailed Pre-Flight Checklist:

1. **Mission Planning:** Define objectives, review area, check airspace/regulations, permissions.

References

2. **Weather Check:** Verify conditions are within limits. Check the GPS forecast.
3. **Site Survey:** Check takeoff/landing area, obstructions, interference, and safe zones.
4. **Hardware Inspection (Drone):** Frame, arms, motors, props, wiring, landing gear, payloads, GPS mast.
5. **Hardware Inspection (Ground):** Tx battery/condition, GCS charged/running, telemetry connected.
6. **Battery Check:** Inspect the flight battery and ensure it is fully charged.
7. **Power On Sequence:** Tx ON first, secure the battery and connect the battery to the drone.
8. **System Initialization:** Wait for FC boot, GPS 3D Fix, verify Home position, connect GCS via telemetry, and check data stream/warnings.
9. **Pre-Arm Checks (GCS):** Resolve any "Pre-Arm Check Failed" errors.
10. **Control & Failsafe Test (Props OFF recommended):** Arm, verify control response, test mode switching, test RC failsafe (turn off Tx), disarm.
11. **Payload Check (Final):** Verify camera/gimbal/sprayer respond if applicable.
12. **Final Clear & Takeoff:** Clear area, announce takeoff, arm, brief hover test.

Detailed Post-Flight Checklist:

1. **Landing & Disarm:** Land safely, disarm immediately.
2. **Power Off Sequence:** Disconnect the drone battery FIRST, then turn off Tx.
3. **Battery Handling:** Remove battery, check, set to a storage charge, and store safely.
4. **Hardware Inspection:** Inspect drone/payloads for damage.
5. **Data Management:** Download logs/imagery if needed.
6. **Cleaning:** Clean drone/payloads as needed. Flush sprayer.
7. **Storage:** Store drone/equipment appropriately.
8. **Logbook Entry:** Record flight details.

A.4 LiPo Battery Safety Guidelines

Lithium Polymer batteries require careful handling. **NEVER underestimate the risks.**

- **Charging:**

- **ALWAYS** use a LiPo-specific balance charger.
- **NEVER** charge unattended. Charge in a fire-resistant location.
- Set correct cell count/current (1C recommended).
- **NEVER** charge damaged/puffed batteries.
- Stop immediately if battery swells/smokes/gets hot.

- **Discharging (Use):**

- **NEVER** over-discharge (below 3.0V/cell under load). Use voltage failsafe.
- Avoid physical damage. Inspect after crashes.

- **Storage:**

- Store at storage voltage (3.80-3.85V/cell) in a cool, dry place inside a fire-resistant container (LiPo bag).
- Do not store fully charged/discharged long-term.

- **Handling:**

- **NEVER** short-circuit terminals.
- **NEVER** puncture, deform, crush, or disassemble.

- **Disposal:**

- **NEVER** put in regular trash. Discharge fully, soak in salt water (outdoors), and take to hazardous waste/recycling facility.

- **Fire Safety:**

- Keep a Class D extinguisher or sand nearby. Water is generally not recommended.

References

A.5 Software and Resource Links

- ArduPilot Firmware (ArduCopter): <https://ardupilot.org/copter/>
- PX4 Firmware: <https://px4.io/>
- Mission Planner GCS: <https://ardupilot.org/planner/>
- QGroundControl GCS: <http://qgroundcontrol.com/>
- Pixhawk Open Hardware Standards: <https://pixhawk.org/>
- Radiolink Website (AT10): <http://www.radiolink.com/>
- Raspberry Pi Foundation: <https://www.raspberrypi.org/>
- Intel Neural Compute Stick: <https://www.intel.com/ncs>
- OpenVINO Toolkit: <https://docs.openvino.ai/>
- eCalc Propeller Calculator: <https://www.ecalc.ch/>

Note: Always refer to manufacturer datasheets and official documentation for the most accurate and up-to-date information.